Highlights

Bayesian Separation of Latent Spatial Fields from Model Bias via Graph Laplacian Eigenmodes

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- Separates latent spatial fields from structured bias using graph-Laplacian modes.
- Recasts additive process–bias confounding as Bayesian modal allocation.
- Quantifies field, bias, and allocation uncertainty in simulation and SST analyses.

Bayesian Separation of Latent Spatial Fields from Model Bias via Graph Laplacian Eigenmodes

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ARTICLE INFO

Keywords:
Bayesian spatial statistics
Model discrepancy
Graph Laplacian
Computer model ensembles
Sea-surface temperature

ABSTRACT

Numerical model ensembles and physical observations provide complementary but imperfect information about spatial environmental processes. We develop a Bayesian method for separating a latent spatial field from structured model bias when the ensemble constrains only their additive mean and observations are available at a subset of locations. The method represents both fields using graph-Laplacian eigenmodes and assigns low-frequency modes to either the process or the bias through latent allocation indicators. This turns a continuously confounded decomposition into a finite allocation problem over spatial scales, yielding posterior summaries for both field uncertainty and process–bias attribution. A controlled ocean-transport simulation shows that the method recovers coherent large-scale discrepancy while preserving accurate reconstruction of the latent field. An application to January 2010 Pacific sea-surface temperature combines an observation-based analysis with an E3SM ensemble and produces a smooth latent temperature field, a basin-scale bias correction, and interpretable allocation probabilities over spatial modes. The results show that modal allocation provides a practical Bayesian framework for process–bias separation in partially identified spatial problems.

1. Introduction

Numerical models and observation systems provide complementary but imperfect views of complex physical systems. In many environmental and geophysical problems, one has incomplete and noisy observations of the process of interest together with spatially complete output from one or more numerical models. A central statistical task is then to combine these sources of information without treating the numerical model as an unbiased surrogate for nature. This perspective is standard in Bayesian data assimilation, model validation, and the broader literature on inference for physics-based simulators (Wikle and Berliner, 2007; Bayarri, Paulo, Berger, Sacks, Cafeo, Cavendish, Lin and Tu, 2007; Goldstein and Rougier, 2009; Brynjarsdóttir and O'Hagan, 2014).

The main difficulty is model discrepancy. When a numerical model omits or misrepresents relevant physics, the gap between simulator output and reality is itself a structured stochastic object. If that discrepancy is ignored, inference can become biased and overconfident; if it is modeled too flexibly, it can become confounded with the latent process one is trying to recover (Kennedy and O'Hagan, 2001; Bayarri et al., 2007; Higdon, Gattiker, Williams and Rightley, 2008; Brynjarsdóttir and O'Hagan, 2014). In the spatial setting considered here, the issue is especially acute because the observational data inform only a subset of the domain, while the computer-model ensemble informs a spatially complete field shifted by an unknown bias field. The problem is therefore not ordinary smoothing or interpolation. It is a partially identified decomposition problem: which spatial features of the model ensemble should be attributed to the latent process, and which should be attributed to systematic model error?

This question lies at the intersection of two ideas in modern spatial statistics. The first is that large spatial problems often benefit from basis representations that reduce dimension while preserving interpretable spatial structure (Cressie and Johannesson, 2008). The second is that many effective spatial priors are defined through local neighborhood relations or sparse precision operators, rather than through dense covariance matrices (Besag, 1974; Rue and Held, 2005; Lindgren, Rue and Lindström, 2011). The present paper combines these perspectives by working in a basis induced by the graph of the spatial lattice itself.

In this work we use eigenvectors of the graph Laplacian as an ordered collection of spatial modes. From the viewpoint of graph signal processing, these eigenvectors play the role of Fourier modes on an irregular domain, and their eigenvalues provide a natural ordering from smooth, large-scale variation to rough, fine-scale variation (Shuman,

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Narang, Frossard, Ortega and Vandergheynst, 2013; Ortega, Frossard, Kovacevic, Moura and Vandergheynst, 2018). This ordering is useful for discrepancy modeling because structured model bias is usually expected to be spatially coherent, whereas unrestricted high-frequency discrepancy can easily become a residual noise surface. The Laplacian basis therefore allows the process–bias separation to be expressed in terms of spatial scale rather than in terms of an arbitrary orthogonal rotation of the parameter space.

Our goal is to solve the following unidentifiability problem. With $\boldsymbol{X}$ denoting the underlying process and $\boldsymbol{\beta}$ denoting model discrepancy, we assume that the computer-model output constrains only the additive term $\boldsymbol{X} + \boldsymbol{\beta}$, so the two components are not separately identifiable without additional structure. Additional information, albeit quite limited, about $\boldsymbol{X}$ is available via observational data. Our goal is to disentangle, as much as possible, the two components. We remove the continuous confounding by modeling $\boldsymbol{X}$ and $\boldsymbol{\beta}$ to occupy complementary subsets of graph-Laplacian modes. We take a Bayesian approach and aim to explore a posterior distribution which will now contain not only information about $\boldsymbol{X}$ and $\boldsymbol{\beta}$, but also information about the spatial scales assigned to the signal and to structured discrepancy.

A related but distinct use of orthogonality appears in projection methods for inverse problems with nuisance components. For example, Calvetti, Hyvönen, Kolehmainen and Somersalo (2025) study linear inverse problems in which orthogonal projections are used to remove or attenuate nuisance contributions in the data. That viewpoint is useful here because it emphasizes orthogonality as a way to control confounding. In contrast, rather than treating model discrepancy as a nuisance component, the bias field is retained as a scientific target, and posterior uncertainty about its modal support is part of the inferential output. Similarly, Moran-eigenvector spatial filtering uses eigenvectors of centered spatial-weights operators as regressors or filters to reduce residual spatial autocorrelation in regression models (Moran, 1950; Griffith, 2003; Dray, Legendre and Peres-Neto, 2006; Griffith and Peres-Neto, 2006; Tiefelsdorf and Griffith, 2007). By contrast, we use graph-Laplacian modes (eigenvectors) ordered by a Dirichlet-energy roughness criterion. Rather than aiming to whiten residuals, the modes define prior coordinates for two latent fields whose sum is observed through the model ensemble data. A mode assigned to $\boldsymbol{\beta}$ is an explicit component of model bias, and a mode assigned to $\boldsymbol{X}$ is an explicit component of the latent process.

The contribution of this paper is to turn this idea into a Bayesian modal-allocation model. We represent the latent process $\boldsymbol{X}$ and the bias field $\boldsymbol{\beta}$ on complementary subsets of Laplacian modes, select graph-frequency-weighted Gaussian priors on their spectral coefficients, and update the allocation indicators in a collapsed Gibbs sampler. The resulting posterior summarizes both field and allocation uncertainty. In applications, this produces an interpretable decomposition of the additive model mean and a diagnostic record of which spatial scales are, and are not, separable from the available data.

The remainder of the paper is organized as follows. Section 2 reviews the lattice graph and Laplacian preliminaries. Section 3 develops the spectral re-parameterization, prior specification, posterior distributions, and Markov chain Monte Carlo algorithm. Section 4 evaluates the method in a controlled simulation study where both the latent field and the induced discrepancy are known. Section 5 presents an application to Pacific sea-surface temperature, illustrating how the proposed decomposition separates a latent temperature field from a spatially organized climate-model bias. Section 6 concludes with a discussion of interpretation, limitations, and connections to existing spectral approaches.

2. Preliminaries

Throughout the manuscript we assume that all processes appearing below are "vectorized", i.e., they live in a finite-dimensional Euclidean space. For any given application, this is no trivial task in general; however, the focus of this manuscript is to assess the model-process discrepancy at the finite dimensional level. Let $\boldsymbol{X} \in \mathbb{R}^S$ denote a latent, unobserved field (i.e., "nature", or "truth") on a spatial lattice with S grid sites, and let $\boldsymbol{Y}^o \in \mathbb{R}^N$ denote the physical observations at $N \ll S$ observed grid sites. Suppose further that a computer model provides an ensemble of spatial fields $\boldsymbol{Y}_1^c, \dots, \boldsymbol{Y}_M^c \in \mathbb{R}^S$. We consider the Class II model setup of Berliner, Herbei, Wikle and Milliff (2023), written as

$$\boldsymbol{Y}^o \mid \boldsymbol{X}, \sigma_{yo}^2 \sim \mathcal{N}(\mathcal{H}(\boldsymbol{X}), \sigma_{yo}^2 \boldsymbol{I}_N), \tag{1}$$

$$\boldsymbol{Y}_i^c \mid \boldsymbol{X}, \boldsymbol{\beta}, \sigma_{yc}^2 \overset{\text{ind}}{\sim} \mathcal{N}\big(\mathcal{G}(\boldsymbol{X}) + \boldsymbol{\beta}, \sigma_{yc}^2 \boldsymbol{I}_S\big), \qquad i = 1, \dots, M\ , \tag{2}$$

where the vector $\boldsymbol{\beta} \in \mathbb{R}^S$ is viewed as the *computer model bias*. Above, $\mathcal{H}(\boldsymbol{X})$ and $\mathcal{G}(\boldsymbol{X})$ represent dimension-reduced transformations of the process $\boldsymbol{X}$. For example, they can be proxies for a sub-sampling procedure, or a local averaging smoother. In the setup considered below, we take $\mathcal{H}$ to be a *sub-sampler*, i.e., $\mathcal{H}(\boldsymbol{X})$ is a sub-vector of $\boldsymbol{X}$, of the same dimension as $\boldsymbol{Y}^o$, while $\mathcal{G}(\boldsymbol{X})$ remains the identity operator. These simplifications will aid with computing in the applications presented later, but general forms of $\mathcal{G}, \mathcal{H}$ can be incorporated in the methodology described below, with appropriate computational modifications. Our stated goal is to learn the separation between $\boldsymbol{X}$ and $\boldsymbol{\beta}$, in a Bayesian framework. The variances $\sigma^2_{yo}, \sigma^2_{yc}$ are assumed unknown and will also be estimated. Going forward, we remove $\mathcal{G}$ from our analysis and $\mathcal{H}$ is taken to be a $N \times S$ matrix, thus we replace $\mathcal{H}(\boldsymbol{X})$ by $\mathcal{H}\boldsymbol{X}$ and treat this as ordinary matrix multiplication.

The statistical difficulty is immediate. The computer-model output informs only the sum $\boldsymbol{X} + \boldsymbol{\beta}$, whereas the observations inform only the sampled components of $\boldsymbol{X}$. Even with abundant observational data (N large) this is generally insufficient to fully "separate" the latent process $\boldsymbol{X}$ from the model error $\boldsymbol{\beta}$. Without additional structure, the decomposition of the computer-model mean into process and bias components is not identifiable, since many pairs $(\boldsymbol{X}, \boldsymbol{\beta})$ can yield the same sum. The purpose of the spectral construction developed below is to regularize this decomposition in a way that is directly tied to the geometry of the spatial lattice and to the roughness properties of the two latent fields.

First, we assume that all processes live in a rectangular spatial domain Ω which is endowed with a regular lattice with R rows and C columns, so that $S = RC$. This is helpful from a computational perspective, but in general it is not a limiting factor, as we shall see in Section 5. Grid sites are indexed either by a single index $s \in \{1, \dots, S\}$ or by a pair (r, c) with $r \in \{1, \dots, R\}$ and $c \in \{1, \dots, C\}$. This is sufficient, and we do not explicitly use the geographical latitude-longitude coordinates, although in applications this can be done immediately. We use column-major ordering, $s = r + (c-1)R$.

We view the lattice of sites under consideration as a graph with vertices at locations indexed by s. Undirected edges are defined by the north-south-east-west adjacency on the lattice and we use $s \sim s'$ to indicate that sites s, s' are neighbors. Let d_s be the number of neighbors of site s. The graph Laplacian $\boldsymbol{L} = (L_{ss'}) \in \mathbb{R}^{S \times S}$ is defined by

$$L_{ss'} = \begin{cases} d_s, & s = s', \\ -1, & s \sim s', \\ 0, & \text{otherwise.} \end{cases}$$

The matrix $\boldsymbol{L}$ is symmetric and positive semidefinite, as can be seen from the identity

$$\boldsymbol{f}^\top \boldsymbol{L} \boldsymbol{f} = \sum_{s \sim s'} (f_s - f_{s'})^2, \qquad \boldsymbol{f} \in \mathbb{R}^S. \tag{3}$$

Because the lattice graph is connected, $\boldsymbol{L}$ has a one-dimensional null space spanned by the constant vector $\mathbf{1}$. Hence, its smallest eigenvalue is $\lambda_1 = 0$ with eigenvector proportional to $\mathbf{1} = (1, 1, \dots, 1)^\top$. Writing the orthonormal eigendecomposition as

$$\boldsymbol{L} = \boldsymbol{U} \boldsymbol{\Lambda} \boldsymbol{U}^\top, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_S), \tag{4}$$

with $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_S$, each eigenvector $\boldsymbol{u}_k$ represents a spatial mode and each eigenvalue λ_k quantifies *the roughness* of that mode. It is therefore natural to interpret low-index modes as large-scale spatial variation and high-index modes as fine-scale variation. The spectral ordering supplied by the graph Laplacian is the basis for the re-parameterization which follows in Section 3.

There is a well-known connection between graphs, their Laplacians and the theory of Gaussian Markov Random Fields (GMRF) (Besag, 1974; Rue and Held, 2005). For $\delta > 0$ and $\delta_0 > 0$, define the precision matrix

$$\boldsymbol{Q}(\delta, \delta_0) = \delta \boldsymbol{L} + \delta_0 \boldsymbol{I}_S. \tag{5}$$

A zero-mean Gaussian distribution with precision matrix $\boldsymbol{Q}(\delta, \delta_0)$ is a proper Gaussian Markov random field on the lattice defined by $\boldsymbol{L}$. The parameter δ controls the penalty on local differences, and thus the *smoothness* of the random field. The parameter $\delta_0 > 0$ ensures that $\boldsymbol{Q}(\delta, \delta_0)$ is invertible, and it defines a proper zero-mean Gaussian distribution (Besag, 1974; Rue and Held, 2005). From (4), we see that

$$\boldsymbol{Q}(\delta, \delta_0) = \boldsymbol{U} \left(\delta \boldsymbol{\Lambda} + \delta_0 \boldsymbol{I}_S \right) \boldsymbol{U}^\top. \tag{6}$$

Consequently, the following relations hold. They will be useful in the computations that follow:

$$\boldsymbol{Q}(\delta,\delta_0)^{-1} = \boldsymbol{U}\left(\delta\boldsymbol{\Lambda} + \delta_0 \boldsymbol{I}_S\right)^{-1}\boldsymbol{U}^\top, \qquad |\boldsymbol{Q}(\delta,\delta_0)| = \prod_{k=1}^{S}(\delta\lambda_k + \delta_0),$$

where $|\cdot|$ denotes the determinant. If $\boldsymbol{Z} \sim \mathcal{N}\left(\boldsymbol{0}, \boldsymbol{Q}(\delta,\delta_0)^{-1}\right)$ and $\boldsymbol{\alpha} = \boldsymbol{U}^\top \boldsymbol{Z} = (\alpha_k)$, then the coefficients α_k are independent Gaussian variables with variances $(\delta\lambda_k + \delta_0)^{-1}$. As such, large frequencies (large λ_k) induce a shrinkage effect on α_k (smaller variances).

On a full $R \times C$ rectangular lattice, the graph Laplacian has a Kronecker-sum representation. Let $\boldsymbol{L}_R$ denote the path-graph Laplacian on R nodes and let $\boldsymbol{L}_C$ denote the path-graph Laplacian on C nodes. Then

$$\boldsymbol{L} = \boldsymbol{I}_C \otimes \boldsymbol{L}_R + \boldsymbol{L}_C \otimes \boldsymbol{I}_R. \tag{7}$$

If $(\lambda_p^{(R)}, \boldsymbol{u}_p^{(R)})$ and $(\lambda_q^{(C)}, \boldsymbol{u}_q^{(C)})$ are the eigenpairs of $\boldsymbol{L}_R$ and $\boldsymbol{L}_C$, then

$$\lambda_{pq} = \lambda_p^{(R)} + \lambda_q^{(C)}, \qquad \boldsymbol{u}_{pq} = \boldsymbol{u}_q^{(C)} \otimes \boldsymbol{u}_p^{(R)} \tag{8}$$

are the eigenpairs of $\boldsymbol{L}$ (Chung, 1997). For a path graph of length n, one may take the discrete cosine basis

$$\lambda_k^{(n)} = 2 - 2\cos\left(\frac{\pi(k-1)}{n}\right), \tag{9}$$

with eigenvectors

$$u_k^{(n)}(i) = \begin{cases} n^{-1/2}, & k = 1, \\ \sqrt{\frac{2}{n}}\cos\left(\frac{\pi(k-1)(i-1/2)}{n}\right), & k = 2, \dots, n, \end{cases} \tag{10}$$

which is the standard discrete cosine transform basis (Ahmed, Natarajan and Rao, 1974).

The computational consequence is important. Multiplication by $\boldsymbol{U}$ or $\boldsymbol{U}^\top$ need not be implemented as a dense matrix operation. Instead, on a regular grid the required projections can be computed by separable discrete cosine transforms, which reduces the cost from quadratic order in S to essentially linear-logarithmic order on large lattices. For lattices of moderate size, a one-time explicit eigendecomposition is feasible. For larger grids, however, the regular-grid structure provides a direct route to fast preprocessing and fast repeated evaluation of spectral projections.

3. Spectral re-parameterization using Laplacian eigenmodes

As discussed, the additive mean $\boldsymbol{X} + \boldsymbol{\beta}$ in (2) is not identifiable without further restrictions. If the two latent fields are allowed to occupy the same function space, then any common perturbation can be added to $\boldsymbol{X}$ and subtracted from $\boldsymbol{\beta}$ without changing the computer-model mean. To remove this continuous confounding, we restrict the two fields to orthogonal subspaces of the Laplacian basis. This idea is related in spirit to restricted spatial regression, where overlap between fixed and spatial effects is removed by projection onto orthogonal subspaces (Reich, Hodges and Zadnik, 2006; Hughes and Haran, 2013); here the competing objects are two latent spatial fields rather than a regression effect and a spatial random effect.

Our strategy is to construct a re-parameterization of the two fields, using the Laplacian basis. We introduce random allocation indicators $B_k \in \{0, 1\}$, $k = 1, 2, \dots, S$, which are interpreted as follows:

$$B_k = 1 \iff \text{mode } k \text{ is assigned to } \boldsymbol{X}, \qquad B_k = 0 \iff \text{mode } k \text{ is assigned to } \boldsymbol{\beta}.$$

The working principle is that the model bias should exhibit large-scale structure, indicative of an imperfect computer model, which might be missing certain forcings, or structural components. In light of this, we will only allow low-frequency modes to be allocated to $\boldsymbol{\beta}$, thus avoiding the scenario where the model bias might include small-scale noise, which is difficult to interpret and is known to lead to over-fitting.

For that, let $\mathcal{K}_0 = \{2, 3, \dots, K_0 + 1\}$ denote the first K_0 nonconstant Laplacian modes. Only modes in $\mathcal{K}_0$ will be allowed to be allocated to $\boldsymbol{\beta}$, while the remaining modes will always be allocated to $\boldsymbol{X}$. Collect these indicator variables in an S-dimensional vector $\boldsymbol{B}$:

$$\boldsymbol{B} \equiv [B_1, \underbrace{B_2, B_3, B_4, \dots, B_{K_0+1}}_{\text{modes allocated to } \boldsymbol{\beta} \text{ or } \boldsymbol{X}}, \underbrace{B_{K_0+2}, \dots, B_S}_{\text{modes allocated to } \boldsymbol{X}}]^\top \in \mathbb{R}^S . \tag{11}$$

The constant mode is always assigned to the process field, so we set $B_k \equiv 1$ for $k \notin \mathcal{K}_0$ and allow a random $B_k \in \{0, 1\}$ for $k \in \mathcal{K}_0$. The indicators B_k can be used to define the corresponding index sets as

$$\mathcal{I}_\beta(\boldsymbol{B}) = \{k \in \mathcal{K}_0 \,:\, B_k = 0\}, \qquad \mathcal{I}_X(\boldsymbol{B}) = \{1, \dots, S\} \setminus \mathcal{I}_\beta(\boldsymbol{B}), \tag{12}$$

and let $d_\beta = |\mathcal{I}_\beta(\boldsymbol{B})|$ and $d_X = |\mathcal{I}_X(\boldsymbol{B})|$ denote the sizes of these sets. Collect the modes corresponding to $\boldsymbol{X}$ and $\boldsymbol{\beta}$ in separate eigenvector matrices

$$\boldsymbol{U}_\beta = \big[\boldsymbol{u}_k \,:\, k \in \mathcal{I}_\beta(\boldsymbol{B})\big], \qquad \boldsymbol{U}_X = \big[\boldsymbol{u}_k \,:\, k \in \mathcal{I}_X(\boldsymbol{B})\big]. \tag{13}$$

Because $\boldsymbol{U}$ is orthonormal, the column spaces of $\boldsymbol{U}_\beta$ and $\boldsymbol{U}_X$ are orthogonal complements:

$$\boldsymbol{U}_\beta^\top \boldsymbol{U}_X = \boldsymbol{0}, \qquad \boldsymbol{U}_\beta \boldsymbol{U}_\beta^\top + \boldsymbol{U}_X \boldsymbol{U}_X^\top = \boldsymbol{I}_S. \tag{14}$$

Having introduced the necessary notation, we can now introduce the re-parameterization. Define coefficient vectors $\boldsymbol{b} \in \mathbb{R}^{d_\beta}$ and $\boldsymbol{x} \in \mathbb{R}^{d_X}$, and, given an allocation vector $\boldsymbol{B}$, define $\boldsymbol{\beta}$ and $\boldsymbol{X}$ as

$$\boldsymbol{X} = \boldsymbol{U}_X \boldsymbol{x} \in \mathbb{R}^S, \qquad \boldsymbol{\beta} = \boldsymbol{U}_\beta \boldsymbol{b} \in \mathbb{R}^S. \tag{15}$$

Equation (15) is at the core of our approach. In conjunction with (14), it implies the deterministic orthogonality constraint

$$\boldsymbol{\beta}^\top \boldsymbol{X} = \boldsymbol{b}^\top \boldsymbol{U}_\beta^\top \boldsymbol{U}_X \boldsymbol{x} = 0.$$

The continuous non-identifiability of the additive decomposition is thereby removed. The remaining uncertainty concerns only the discrete assignment of low-frequency modes between $\boldsymbol{X}$ and $\boldsymbol{\beta}$, which can be summarized by the posterior distribution of $\boldsymbol{B}$; in that sense the problem is transformed from continuous confounding to a finite partially identified allocation problem (Gustafson, 2015).

The re-parameterization in (15) no longer determines proper GMRF distributions on $\boldsymbol{X}$ and $\boldsymbol{\beta}$ on the original lattice. Conditionally on $\boldsymbol{B}$, the fields $\boldsymbol{X} = \boldsymbol{U}_X \boldsymbol{x}$ and $\boldsymbol{\beta} = \boldsymbol{U}_\beta \boldsymbol{b}$ are Gaussian fields supported on complementary subspaces, but their precision matrices are generally no longer sparse. The role of this construction is different: to regularize the non-identifiable decomposition $\boldsymbol{X} + \boldsymbol{\beta}$ in directions that are directly tied to the geometry of the original lattice and to spatial scale. By allocating complementary sets of basis modes to the two fields, we make precise what aspect of the sum is being attributed to the latent process and what aspect is being attributed to model discrepancy.

The graph Laplacian basis is a natural vehicle for this construction because its eigenvectors are the graph analogue of Fourier modes. If $\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, then the eigenvalue λ_k attached to $\boldsymbol{u}_k$ quantifies the roughness of that mode. The ordering of the eigenvalues gives a concrete interpretation to the allocation vector $\boldsymbol{B}$: assigning a mode to $\boldsymbol{\beta}$ means that the corresponding spatial scale is regarded as potentially attributable to structured model error, while assigning it to $\boldsymbol{X}$ means that the same scale is attributed to the latent process. This is the substantive modeling choice. The goal is therefore to choose subspaces whose coordinates have a defined spatial meaning. In a generic basis, orthogonality is only an algebraic convenience: it does not correspond to smoothness on the lattice, to a penalty on local differences, or to any interpretable notion of large- versus small-scale structure. By contrast, the Laplacian basis is singled out by the spatial graph itself, so the decomposition is anchored in the domain geometry rather than in an arbitrary coordinate system.

3.1. Re-parameterized Bayesian model

We are now in a position to restate the full Bayesian statistical model in terms of the newly re-parametrized fields. We start by selecting the prior distribution on the allocation vector $\boldsymbol{B}$ to be a product Bernoulli law

$$p(\boldsymbol{B}) = \prod_{k \in \mathcal{K}_0} p_k^{B_k} (1 - p_k)^{1 - B_k}, \tag{16}$$

where $p_k \in [0,1]$ are prior probabilities. In the applications discussed later, we find that fixing the values $\{p_k\}$ a priori works very well. As an alternative, the prior probabilities can be linked to the graph frequencies. Conditionally on $\boldsymbol{B}$, we select the following prior distributions for $\boldsymbol{x}, \boldsymbol{b}$. Consider diagonal matrices

$$\boldsymbol{D}_\beta = \mathrm{diag}\big(\delta_1\lambda_k + \delta_{01} \,:\, k \in \mathcal{I}_\beta(\boldsymbol{B})\big), \qquad \boldsymbol{D}_X = \mathrm{diag}\big(\delta_2\lambda_k + \delta_{02} \,:\, k \in \mathcal{I}_X(\boldsymbol{B})\big), \tag{17}$$

where $\delta_1, \delta_{01}, \delta_2, \delta_{02}$ are specified, positive values. We select conditionally independent prior distributions:

$$\boldsymbol{b} \mid \boldsymbol{B} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{D}_\beta^{-1}), \qquad \boldsymbol{x} \mid \boldsymbol{B} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{D}_X^{-1}), \qquad \boldsymbol{b} \perp\!\!\!\perp \boldsymbol{x} \mid \boldsymbol{B}. \tag{18}$$

The choice of the diagonal precision matrices $\boldsymbol{D}_\beta$ and $\boldsymbol{D}_X$ is also deliberate. Rough modes (large frequencies λ_k) are shrunk more strongly than smoother modes, in a way that is monotone, graph-calibrated and parsimonious. The parameters $\delta_1, \delta_2 > 0$ control the overall strength of roughness penalization, while $\delta_{01}, \delta_{02} > 0$ act as safeguards against the lowest-frequency modes becoming singular. It is clear that arbitrary diagonal matrices could also be used here, but they would encode a separate variance choice for each retained mode. That is a much less structured prior specification: it introduces many more degrees of freedom, it is highly basis-dependent, and it no longer corresponds to a coherent smoothness law on the lattice. Unless there is strong external scientific information favoring specific modes, such a prior would hard-code mode-by-mode preferences that are difficult to justify and harder to interpret. The form $\delta\lambda_k + \delta_0$ avoids this problem by tying shrinkage to a single physically meaningful ordering of the spectrum. It therefore serves as a default regularization device directly connected to the geometry of the spatial domain.

With the notation introduced above, the re-parameterized data model is therefore

$$\boldsymbol{Y}^o \mid \boldsymbol{x}, \boldsymbol{B}, \sigma_{yo}^2 \sim \mathcal{N}(\mathcal{H}\boldsymbol{U}_X\boldsymbol{x}, \sigma_{yo}^2\boldsymbol{I}_N), \tag{19}$$

$$\boldsymbol{Y}_i^c \mid \boldsymbol{b}, \boldsymbol{x}, \boldsymbol{B}, \sigma_{yc}^2 \overset{\text{ind}}{\sim} \mathcal{N}(\boldsymbol{U}_\beta\boldsymbol{b} + \boldsymbol{U}_X\boldsymbol{x}, \sigma_{yc}^2\boldsymbol{I}_S), \qquad i = 1, \dots, M. \tag{20}$$

For the ensemble data, two sufficient statistics play a central role:

$$\overline{\boldsymbol{Y}}^c = \frac{1}{M}\sum_{i=1}^M \boldsymbol{Y}_i^c, \qquad \boldsymbol{W}^c = \sum_{i=1}^M \|\boldsymbol{Y}_i^c - \overline{\boldsymbol{Y}}^c\|^2. \tag{21}$$

As such, we have

$$\overline{\boldsymbol{Y}}^c \mid \boldsymbol{b}, \boldsymbol{x}, \boldsymbol{B}, \sigma_{yc}^2 \sim \mathcal{N}\left(\boldsymbol{U}_\beta\boldsymbol{b} + \boldsymbol{U}_X\boldsymbol{x}, \frac{\sigma_{yc}^2}{M}\boldsymbol{I}_S\right), \tag{22}$$

and, for any $\boldsymbol{\mu} \in \mathbb{R}^S$,

$$\sum_{i=1}^M \|\boldsymbol{Y}_i^c - \boldsymbol{\mu}\|^2 = \boldsymbol{W}^c + M\|\overline{\boldsymbol{Y}}^c - \boldsymbol{\mu}\|^2. \tag{23}$$

Equation (23) shows that, conditional on σ_{yc}^2, inference for the mean structure of the computer model depends on the ensemble only through $\overline{\boldsymbol{Y}}^c$, whereas $\boldsymbol{W}^c$ is the additional term required to learn σ_{yc}^2. It will be useful to make the following rearrangement of eigenvectors, conditionally on $\boldsymbol{B}$:

$$\widetilde{\boldsymbol{U}} = \big[\boldsymbol{U}_\beta\ \boldsymbol{U}_X\big], \qquad \widetilde{\boldsymbol{U}}^\top\widetilde{\boldsymbol{U}} = \boldsymbol{I}_S,$$

and define the transformed computer model information as

$$\overline{\boldsymbol{Z}}^c = \widetilde{\boldsymbol{U}}^\top\overline{\boldsymbol{Y}}^c = \begin{bmatrix}\overline{\boldsymbol{Z}}_\beta^c \\ \overline{\boldsymbol{Z}}_X^c\end{bmatrix}. \tag{24}$$

Then

$$\overline{\boldsymbol{Z}}_\beta^c \mid \boldsymbol{b}, \boldsymbol{B}, \sigma_{yc}^2 \sim \mathcal{N}\left(\boldsymbol{b}, \frac{\sigma_{yc}^2}{M}\boldsymbol{I}_{d_\beta}\right), \qquad \overline{\boldsymbol{Z}}_X^c \mid \boldsymbol{x}, \boldsymbol{B}, \sigma_{yc}^2 \sim \mathcal{N}\left(\boldsymbol{x}, \frac{\sigma_{yc}^2}{M}\boldsymbol{I}_{d_X}\right), \tag{25}$$

with conditional independence between the two blocks given $(\boldsymbol{b}, \boldsymbol{x}, \boldsymbol{B}, \sigma_{yc}^2)$. The computer-model likelihood is therefore diagonalized in the spectral basis, whereas the physical observation likelihood remains coupled through the sampling operator $\mathcal{H}$. The Bayesian specification is completed by selecting Inverse-Gamma priors

$$\sigma_{yo}^2 \sim \mathrm{IG}(\gamma_o, \xi_o), \qquad \sigma_{yc}^2 \sim \mathrm{IG}(\gamma_c, \xi_c), \tag{26}$$

where $\mathrm{IG}(a, b)$ denotes the distribution with density proportional to $x^{-a-1}\exp(-b/x)$ for $x > 0$. Defining $\boldsymbol{\theta} = (\boldsymbol{b}, \boldsymbol{x}, \sigma_{yo}^2, \sigma_{yc}^2)$ and noting that $\|\boldsymbol{Y}^c - \boldsymbol{U}_\beta \boldsymbol{b} - \boldsymbol{U}_X \boldsymbol{x}\|^2 = \|\overline{\boldsymbol{Z}}_\beta^c - \boldsymbol{b}\|^2 + \|\overline{\boldsymbol{Z}}_X^c - \boldsymbol{x}\|^2$, we can write

$$\begin{aligned}\pi(\boldsymbol{\theta} \mid \boldsymbol{B}, \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}) \propto\ & |\boldsymbol{D}_\beta|^{1/2}\, |\boldsymbol{D}_X|^{1/2} \\ & \times \exp\left\{ -\frac{1}{2}\boldsymbol{b}^\top \boldsymbol{D}_\beta \boldsymbol{b} - \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{D}_X \boldsymbol{x} - \frac{1}{2\sigma_{yo}^2}\|\boldsymbol{Y}^o - \mathcal{H}\boldsymbol{U}_X \boldsymbol{x}\|^2 \right\} \\ & \times \exp\left\{ -\frac{1}{2\sigma_{yc}^2}\left[\boldsymbol{W}_c + M\|\overline{\boldsymbol{Z}}_\beta^c - \boldsymbol{b}\|^2 + M\|\overline{\boldsymbol{Z}}_X^c - \boldsymbol{x}\|^2 \right] \right\} \\ & \times \exp\left\{ -\frac{\xi_o}{\sigma_{yo}^2} - \frac{\xi_c}{\sigma_{yc}^2} \right\} (\sigma_{yo}^2)^{-(\gamma_o + N/2+1)} (\sigma_{yc}^2)^{-(\gamma_c + MS/2+1)}. \end{aligned} \tag{27}$$

Finally, the full posterior distribution is the mixed discrete–continuous law

$$\pi(\boldsymbol{B}, \boldsymbol{\theta} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}) \propto p(\boldsymbol{B} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})\, \pi(\boldsymbol{\theta} \mid \boldsymbol{B}, \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})\,.$$

3.2. Posterior marginals

Equation (27) can be used to derive posterior marginal distributions for $\boldsymbol{b}, \boldsymbol{x}$ as follows.

Posterior marginal for $\boldsymbol{b}$. Conditionally on $(\boldsymbol{B}, \sigma_{yc}^2)$, the transformed ensemble block $\overline{\boldsymbol{Z}}_\beta^c$ and the prior (18) yield

$$p(\boldsymbol{b} \mid \boldsymbol{Y}^c_{1:M}, \boldsymbol{B}, \sigma_{yc}^2) \propto \exp\left\{ -\frac{1}{2}\left[M\sigma_{yc}^{-2}\|\overline{\boldsymbol{Z}}_\beta^c - \boldsymbol{b}\|^2 + \boldsymbol{b}^\top \boldsymbol{D}_\beta \boldsymbol{b} \right] \right\}.$$

Completing the square gives

$$\boldsymbol{b} \mid \boldsymbol{Y}^c_{1:M}, \boldsymbol{B}, \sigma_{yc}^2 \sim \mathcal{N}(\boldsymbol{m}_b, \boldsymbol{P}_b^{-1}), \qquad \boldsymbol{P}_b = \boldsymbol{D}_\beta + M\sigma_{yc}^{-2}\boldsymbol{I}_{d_\beta}, \qquad \boldsymbol{m}_b = \boldsymbol{P}_b^{-1} M \sigma_{yc}^{-2} \overline{\boldsymbol{Z}}_\beta^c. \tag{28}$$

We note that, since $\boldsymbol{D}_\beta$ is diagonal, so is $\boldsymbol{P}_b$, and this will be helpful at the computer implementation stage.

Posterior marginal for $\boldsymbol{x}$. Using (18), (19), and (25), we find that

$$p(\boldsymbol{x} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}, \boldsymbol{B}, \sigma_{yo}^2, \sigma_{yc}^2) \propto \exp\left\{ -\frac{1}{2}\left[M\sigma_{yc}^{-2}\|\overline{\boldsymbol{Z}}_X^c - \boldsymbol{x}\|^2 + \sigma_{yo}^{-2}\|\boldsymbol{Y}^o - \mathcal{H}\boldsymbol{U}_X\boldsymbol{x}\|^2 + \boldsymbol{x}^\top \boldsymbol{D}_X \boldsymbol{x} \right] \right\}.$$

Hence

$$\boldsymbol{x} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}, \boldsymbol{B}, \sigma_{yo}^2, \sigma_{yc}^2 \sim \mathcal{N}(\boldsymbol{m}_x, \boldsymbol{P}_x^{-1}), \tag{29}$$

where

$$\boldsymbol{P}_x = \boldsymbol{D}_X + M\sigma_{yc}^{-2}\boldsymbol{I}_{d_X} + \sigma_{yo}^{-2}\boldsymbol{U}_X^\top \mathcal{H}^\top \mathcal{H}\boldsymbol{U}_X, \qquad \boldsymbol{m}_x = \boldsymbol{P}_x^{-1}\left(M\sigma_{yc}^{-2}\overline{\boldsymbol{Z}}_X^c + \sigma_{yo}^{-2}\boldsymbol{U}_X^\top \mathcal{H}^\top \boldsymbol{Y}^o \right). \tag{30}$$

The precision matrix is no longer diagonal because the observation operator couples the retained modes. In practice, inverting $\boldsymbol{P}_x$ can be done using a Cholesky factorization.

Posterior marginals for σ_{yo}^2 *and* σ_{yc}^2. Conditionally on $(\boldsymbol{b}, \boldsymbol{x}, \boldsymbol{B})$, the two variance updates are conjugate:

$$\sigma_{yo}^2 \mid \boldsymbol{x}, \boldsymbol{B}, \boldsymbol{Y}^o \sim \mathrm{IG}\left(\gamma_o + \frac{N}{2}\,,\ \xi_o + \frac{1}{2}\|\boldsymbol{Y}^o - \mathcal{H}\boldsymbol{U}_X\boldsymbol{x}\|^2 \right), \tag{31}$$

$$\sigma_{yc}^2 \mid \boldsymbol{b}, \boldsymbol{x}, \boldsymbol{B}, \boldsymbol{Y}^c_{1:M} \sim \mathrm{IG}\left(\gamma_c + \frac{MS}{2}\,,\ \xi_c + \frac{1}{2}\left[\boldsymbol{W}^c + M\|\overline{\boldsymbol{Y}}^c - \boldsymbol{U}_\beta\boldsymbol{b} - \boldsymbol{U}_X\boldsymbol{x}\|^2 \right] \right). \tag{32}$$

The second update highlights the statistical role of the ensemble. Even when the decomposition of the mean field is uncertain, the within-ensemble term $\boldsymbol{W}^c$ contributes direct information about σ_{yc}^2.

Collapsed posterior marginal for $\boldsymbol{B}$. Although the full posterior distribution of $\boldsymbol{B}$ conditioning on everything else can be written out explicitly, it is advantageous to integrate out $(\boldsymbol{b}, \boldsymbol{x})$. Since $\boldsymbol{W}^c$ is free of $\boldsymbol{B}$ conditional on σ^2_{yc}, the collapsed posterior is

$$p(\boldsymbol{B} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}, \sigma^2_{yo}, \sigma^2_{yc}) \propto p(\boldsymbol{B})\, p(\overline{\boldsymbol{Z}}^c_\beta \mid \boldsymbol{B}, \sigma^2_{yc})\, p(\overline{\boldsymbol{Z}}^c_X, \boldsymbol{Y}^o \mid \boldsymbol{B}, \sigma^2_{yo}, \sigma^2_{yc}). \tag{33}$$

The likelihood contributions in the above posterior can be written out analytically as follows. The second factor above is obtained from (25) and (18):

$$p(\overline{\boldsymbol{Z}}^c_\beta \mid \boldsymbol{B}, \sigma^2_{yc}) = (2\pi)^{-d_\beta/2} \left(\frac{\sigma^2_{yc}}{M}\right)^{-d_\beta/2} \frac{|\boldsymbol{D}_\beta|^{1/2}}{|\boldsymbol{P}_b|^{1/2}} \exp\left\{-\frac{1}{2} M \sigma^{-2}_{yc} \|\overline{\boldsymbol{Z}}^c_\beta\|^2 + \frac{1}{2} \boldsymbol{m}_b^\top \boldsymbol{P}_b \boldsymbol{m}_b\right\}. \tag{34}$$

For the third factor, let

$$\widetilde{\boldsymbol{Y}} = \begin{bmatrix} \overline{\boldsymbol{Z}}^c_X \\ \boldsymbol{Y}^o \end{bmatrix}, \qquad \widetilde{\boldsymbol{D}} = \begin{bmatrix} \boldsymbol{I}_{d_X} \\ \mathcal{H} \boldsymbol{U}_X \end{bmatrix}, \qquad \widetilde{\boldsymbol{R}} = \operatorname{diag}\left(M \sigma^{-2}_{yc} \boldsymbol{I}_{d_X}, \sigma^{-2}_{yo} \boldsymbol{I}_N\right). \tag{35}$$

Then $\widetilde{\boldsymbol{Y}} \mid \boldsymbol{x}, \boldsymbol{B}, \sigma^2_{yo}, \sigma^2_{yc} \sim \mathcal{N}(\widetilde{\boldsymbol{D}}\boldsymbol{x}, \widetilde{\boldsymbol{R}}^{-1})$. Combining this with the prior (18) and then integrating over $\boldsymbol{x}$ yields

$$\begin{aligned} p(\overline{\boldsymbol{Z}}^c_X, \boldsymbol{Y}^o \mid \boldsymbol{B}, \sigma^2_{yo}, \sigma^2_{yc}) = (2\pi)^{-(d_X+N)/2} \left(\frac{\sigma^2_{yc}}{M}\right)^{-d_X/2} (\sigma^2_{yo})^{-N/2} \frac{|\boldsymbol{D}_X|^{1/2}}{|\boldsymbol{P}_x|^{1/2}} \\ \times \exp\left\{-\frac{1}{2} \widetilde{\boldsymbol{Y}}^\top \widetilde{\boldsymbol{R}} \widetilde{\boldsymbol{Y}} + \frac{1}{2} \boldsymbol{m}_x^\top \boldsymbol{P}_x \boldsymbol{m}_x\right\}. \end{aligned} \tag{36}$$

Combining (16), (34), and (36) gives a closed-form expression for the collapsed posterior of the allocation vector. A convenient version is the collapsed log-posterior, up to an additive constant independent of $\boldsymbol{B}$,

$$\begin{aligned} \log p(\boldsymbol{B} \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}, \sigma^2_{yo}, \sigma^2_{yc}) = \sum_{k \in \mathcal{K}_0} \left[B_k \log p_k + (1 - B_k) \log(1 - p_k)\right] \\ + \frac{1}{2} \log |\boldsymbol{D}_\beta| - \frac{1}{2} \log |\boldsymbol{P}_b| + \frac{1}{2} \boldsymbol{m}_b^\top \boldsymbol{P}_b \boldsymbol{m}_b \\ + \frac{1}{2} \log |\boldsymbol{D}_X| - \frac{1}{2} \log |\boldsymbol{P}_x| + \frac{1}{2} \boldsymbol{m}_x^\top \boldsymbol{P}_x \boldsymbol{m}_x. \end{aligned} \tag{37}$$

Therefore, for a single indicator B_k,

$$\mathbb{P}(B_k = 1 \mid \boldsymbol{B}_{-k}, \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M}, \sigma^2_{yo}, \sigma^2_{yc}) = \frac{p_k\, \mathcal{L}(\boldsymbol{B}_{-k}, B_k = 1)}{p_k\, \mathcal{L}(\boldsymbol{B}_{-k}, B_k = 1) + (1 - p_k)\, \mathcal{L}(\boldsymbol{B}_{-k}, B_k = 0)}, \tag{38}$$

where $\mathcal{L}(\boldsymbol{B})$ denotes the product of the two Gaussian marginal likelihood factors in (34) and (36).

3.3. Markov chain Monte Carlo algorithm

The posterior distribution is explored by a Gibbs sampler in which the allocation vector is updated from the collapsed posterior and the continuous variables are updated from their conjugate full conditionals. A single iteration has the following structure.

1. Form the collapsed conditional probabilities (38) for each $k \in \mathcal{K}_0$ and update the indicators sequentially.
2. Given the new allocation vector, reconstruct $\mathcal{I}_\beta(\boldsymbol{B})$, $\mathcal{I}_X(\boldsymbol{B})$, $\boldsymbol{U}_\beta$, $\boldsymbol{U}_X$, $\boldsymbol{D}_\beta$, and $\boldsymbol{D}_X$.
3. Compute $\overline{\boldsymbol{Z}}^c_\beta = \boldsymbol{U}_\beta^\top \overline{\boldsymbol{Y}}^c$ and draw $\boldsymbol{b}$ from (28).
4. Compute $\overline{\boldsymbol{Z}}^c_X = \boldsymbol{U}_X^\top \overline{\boldsymbol{Y}}^c$, factorize $\boldsymbol{P}_x$, and draw $\boldsymbol{x}$ from (29).
5. Draw σ^2_{yo} and σ^2_{yc} from (31) and (32).
6. Reconstruct the spatial fields by $\boldsymbol{\beta} = \boldsymbol{U}_\beta \boldsymbol{b}$ and $\boldsymbol{X} = \boldsymbol{U}_X \boldsymbol{x}$.

This sampler uses the exact conditional distribution of each indicator under the collapsed target, so the discrete update does not require a separate acceptance step. The algorithm is therefore stable, straightforward to implement, and closely aligned with the Gaussian structure of the model.

3.4. Computational strategies

Three implementation features determine the computational efficiency of the method. First, the entire ensemble is reduced to $(\overline{\boldsymbol{Y}}^c, \boldsymbol{W}^c)$ before the Markov chain begins. Forming these sufficient statistics costs $O(MS)$ once. After that point, the per-iteration cost of the algorithm no longer depends on the number of ensemble members. Second, all spectral calculations are organized in the Laplacian basis. On a regular grid this means that the projections $\boldsymbol{U}^\top \overline{\boldsymbol{Y}}^c$ and the extraction of individual modal coefficients can be obtained from a one-time explicit eigendecomposition. The resulting computational savings are especially substantial when repeated projections are required in long Markov chains. Third, the collapsed update of $\boldsymbol{B}$ can be implemented locally. A naive implementation of (38) recomputes the full collapsed objective under both candidate values of B_k and refactorizes a dense matrix of dimension d_X at every coordinate update. If $\bar{d}_X$ denotes a representative dimension of the $\boldsymbol{X}$ block, the cost of one full sweep over $\mathcal{K}_0$ is then of order $O(|\mathcal{K}_0|\bar{d}_X^3)$. The key observation is that toggling a single indicator moves only one eigenmode between $\boldsymbol{U}_X$ and $\boldsymbol{U}_\beta$. Let $k \in \mathcal{K}_0$ and suppose the current state has $B_k = 1$, so that mode k belongs to the process field. Write

$$\boldsymbol{P}_x = \begin{bmatrix} q_k & \boldsymbol{b}_k^\top \\ \boldsymbol{b}_k & \boldsymbol{C}_k \end{bmatrix}, \qquad \boldsymbol{h}_X = \begin{bmatrix} h_k \\ \boldsymbol{h}_r \end{bmatrix}, \tag{39}$$

where $\boldsymbol{h}_X = M\sigma_{yc}^{-2}\overline{\boldsymbol{Z}}_X^c + \sigma_{yo}^{-2}(\mathcal{H}\boldsymbol{U}_X)^\top \boldsymbol{Y}^o$. Define the Schur complement quantities

$$s_k = q_k - \boldsymbol{b}_k^\top \boldsymbol{C}_k^{-1}\boldsymbol{b}_k, \qquad g_k = h_k - \boldsymbol{b}_k^\top \boldsymbol{C}_k^{-1}\boldsymbol{h}_r. \tag{40}$$

Then

$$|\boldsymbol{P}_x| = |\boldsymbol{C}_k|\, s_k, \qquad \boldsymbol{h}_X^\top \boldsymbol{P}_x^{-1}\boldsymbol{h}_X = \boldsymbol{h}_r^\top \boldsymbol{C}_k^{-1}\boldsymbol{h}_r + \frac{g_k^2}{s_k}. \tag{41}$$

These identities imply that the change in the collapsed objective when mode k is moved between the two subspaces can be evaluated by scalar updates and triangular solves with the cached factorization of $\boldsymbol{C}_k$, rather than by a fresh dense factorization of the full matrix $\boldsymbol{P}_x$. The same principle applies when a mode is moved in the opposite direction. The exact conditional probabilities are unchanged, but the dominant cost of one sweep is reduced from cubic to quadratic order in the effective process dimension, namely from $O(|\mathcal{K}_0|\bar{d}_X^3)$ to $O(|\mathcal{K}_0|\bar{d}_X^2)$.

In summary, the computational design rests on a one-time aggregation of the ensemble, spectral diagonalization of the graph-based priors, and local Schur-complement updates for the indicator sweep.

4. A simulation study

The synthetic study follows the oceanographically motivated construction used in the class-II example of Berliner et al. (2023), itself inspired by the inverse problem of McKeague, Nicholls, Speer and Herbei (2005). The physical domain is a rectangular region represented on a 19×37 regular lattice, so that $S = 703$. The observational network consists of the same $N = 335$ grid points used in the source study. The underlying scientific variable is oxygen concentration in a neutral density layer, deep in the South Atlantic Ocean, evolving under advection and diffusion, with a velocity field motivated by large-scale ocean circulation (McKeague et al., 2005).

Because the latent truth is available in this synthetic experiment, the analysis can assess two distinct properties of the method: recovery of the latent field $\boldsymbol{X}$ and recovery of the structured discrepancy field $\boldsymbol{\beta}$. The latent field $\boldsymbol{X}$ is generated from a discretized advection–diffusion system with spatially varying diffusivities, whereas the computer model uses the same velocity field and boundary conditions but replaces those diffusivities by spatially constant values. This misspecification is informative because it arises from a physically meaningful simplification of the transport process. An ensemble of 20 members is generated with the Feynman–Kac simulator described by Herbei and Berliner (2014) and reviewed by Baker, Barbillon, Fadikar, Gramacy, Herbei, Higdon, Huang, Johnson, Ma, Mondal et al. (2022). Figures 1 and 2 show that the ensemble members share a stable basin-scale structure and that the pointwise within-ensemble variance is heterogeneous but modest: the empirical variance has median 0.351, mean 0.448, and maximum 2.492, all small relative to the full oxygen range visible in Figure 1.

The discrepancy induced by the diffusivity misspecification is shown in Figure 3 through the field $\overline{\boldsymbol{Y}}^c - \boldsymbol{X}$. Its dominant features are broad zonal bands and coherent meridional gradients. In transport problems, misspecification of

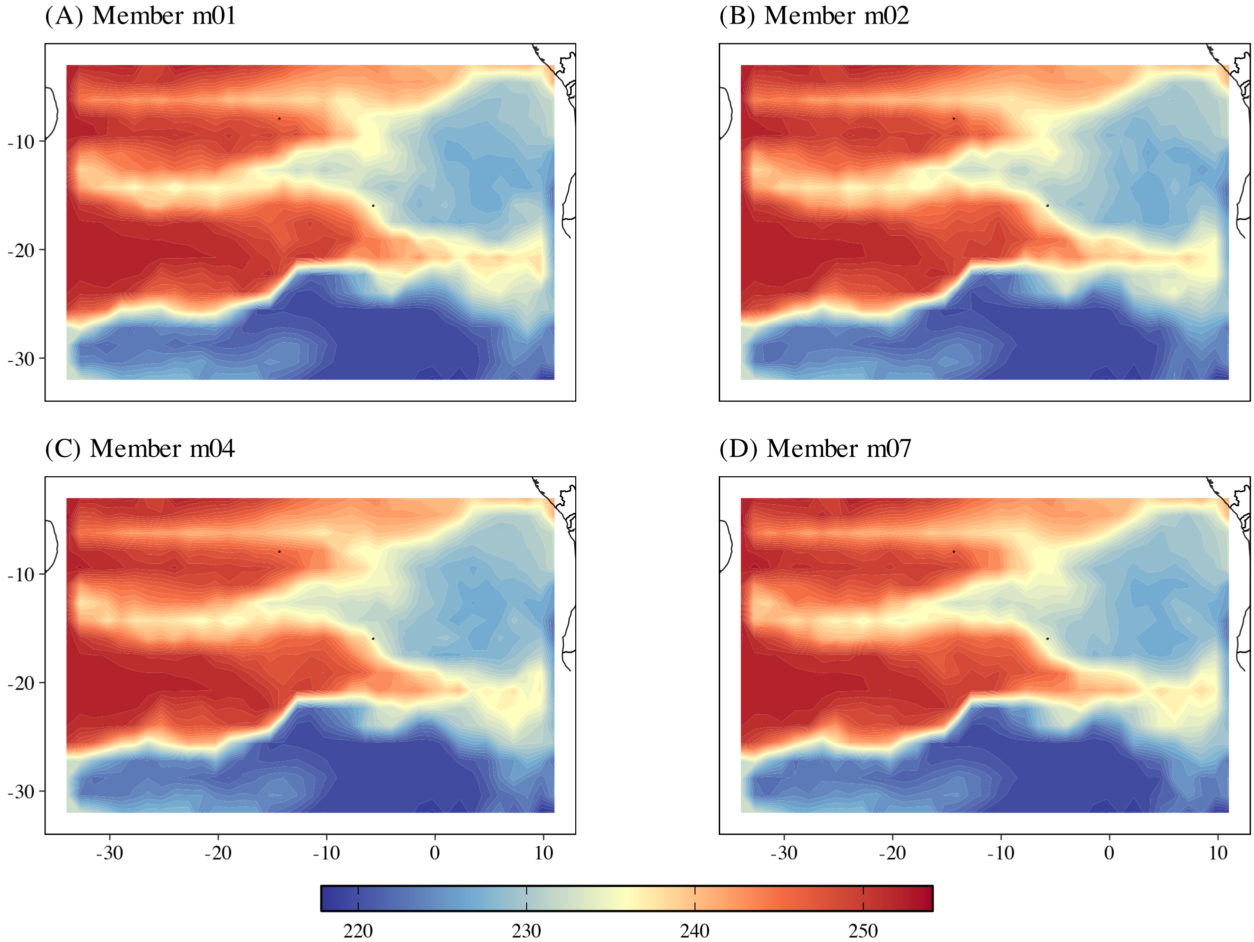

Figure 1: Four representative computer-model ensemble members.

mixing coefficients perturbs tracer distributions over large spatial scales, so the resulting error should concentrate on the smoother part of the spectrum rather than on the highest-frequency modes (Kennedy and O'Hagan, 2001; Higdon et al., 2008; Berliner et al., 2023). The simulation is therefore a direct test of the proposed orthogonal spectral construction: a successful method must recover a latent field close to $\boldsymbol{X}$ while also extracting from the ensemble mean a bias pattern that is spatially organized and physically interpretable.

For this experiment we retain the full Laplacian basis, $K = S = 703$, but only the first $K_0 = 300$ nonconstant modes are eligible for reassignment between $\boldsymbol{X}$ and $\boldsymbol{\beta}$. In preliminary analyses we considered $K_0 \in \{50, 150, 200, 300\}$ and found that $K_0 = 300$ gave the best recovery; we therefore report only the $K_0 = 300$ results. This parameter controls the maximal spatial complexity that can be attributed to model bias. Very small values of K_0 force the discrepancy into an overly smooth subspace, whereas larger values permit the bias field to use additional low- and medium-frequency structure. We set $\delta_{01} = \delta_{02} = 10^{-4}$. The process and bias roughness penalties are held fixed at $\delta_1, \delta_2 \in \{0.5, 0.75, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0\}$. Our simulation grid ranges from equal roughness penalties for $\boldsymbol{X}$ and $\boldsymbol{\beta}$ to settings that impose substantially stronger prior smoothness on the process or bias fields. The Bernoulli allocation probability is fixed to $p_k \in \{0.1, 0.5\}$. In partially identified decompositions, strongly asymmetric choices of p_k should be reserved for situations with clear external scientific information; otherwise a prior-neutral specification is the more defensible default (Gustafson, 2015). We use $\mathrm{IG}(\gamma_* = 4, \xi_* = 0.1)$ priors for σ^2_{yo} and σ^2_{yc}.

For each MCMC run, we compute root-mean-square errors for the posterior mean fields,

$$\mathrm{RMSE}(\boldsymbol{X}) = S^{-1/2}\|\widehat{\boldsymbol{X}} - \boldsymbol{X}\|_2, \qquad \mathrm{RMSE}(\boldsymbol{\beta}) = S^{-1/2}\|\widehat{\boldsymbol{\beta}} - (\bar{\boldsymbol{Y}}^c - \boldsymbol{X})\|_2,$$

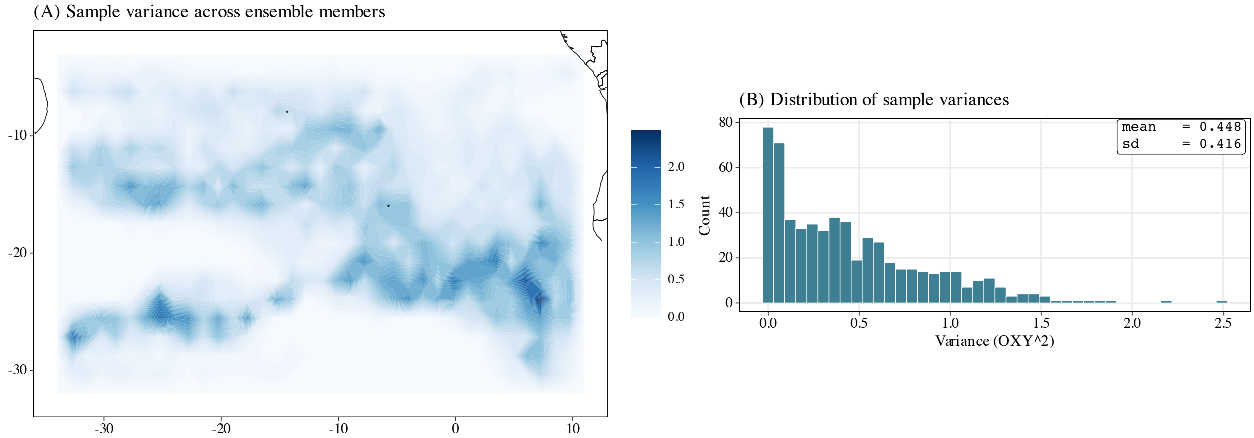

Figure 2: Pointwise sample variances (left) and histogram of sample variances across ensemble members (right). The inset reports the mean and standard deviation of the lattice-wise variance field.

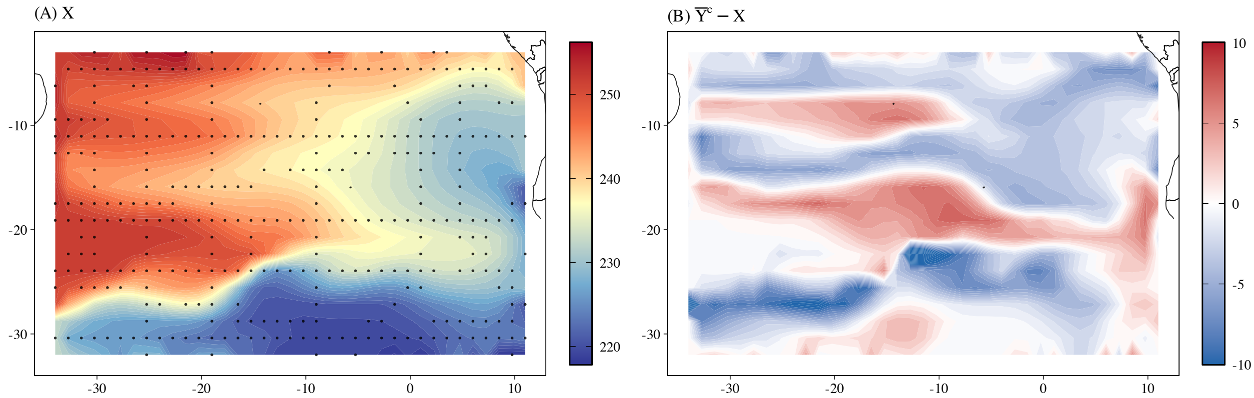

Figure 3: True latent field $\boldsymbol{X}$ (left) and induced discrepancy $\overline{\boldsymbol{Y}}^c - \boldsymbol{X}$ (right). The dots in the left panel indicate locations where observational data are available.

where $\widehat{\boldsymbol{X}}$ and $\widehat{\boldsymbol{\beta}}$ denote posterior means under that run. Table 1 presents the twenty best runs when ranked by RMSE($\boldsymbol{X}$) (first column). We also report the posterior means of $\overline{\sigma^2_{yo}}$, $\overline{\sigma^2_{yc}}$, $\overline{d_\beta}$, the prior smoothness settings δ_1, δ_2, and the (common) prior allocation probability p_k. The fourth column gives the rank with respect to RMSE($\boldsymbol{\beta}$). We select row 2 as a compromise simulation run, as it delivers excellent performance in terms of recovering both the process and bias fields. The subsequent summary figures are based on the settings from row 2 of Table 1.

Table 1: MCMC runs ranked by RMSE($\boldsymbol{X}$) for the simulation study. The fourth column gives the rank with respect to RMSE($\boldsymbol{\beta}$).

Rank	RMSE($\boldsymbol{X}$)	RMSE($\boldsymbol{\beta}$)	Rank($\boldsymbol{\beta}$)	$\overline{\sigma^2_{yo}}$	$\overline{\sigma^2_{yc}}$	p_k	$\overline{d_\beta}$	δ_1	δ_2
1	2.390	3.290	12	5.311	7.275	0.50	136.72	3.00	3.00
2	2.524	2.659	2	4.511	0.617	0.50	144.35	2.50	2.50
3	2.587	3.813	22	6.603	7.699	0.50	96.16	4.00	3.00
4	2.600	2.640	1	5.462	0.487	0.10	245.62	2.00	2.00
5	2.602	2.692	6	4.555	0.488	0.50	145.62	2.00	2.00
6	2.633	2.695	7	4.633	0.430	0.50	145.71	1.50	1.50
7	2.646	2.674	3	5.568	0.431	0.10	245.68	1.50	1.50

Continued on next page

Table 1: MCMC runs ranked by RMSE($\boldsymbol{X}$) for the simulation study, continued.

Rank	RMSE($\boldsymbol{X}$)	RMSE($\boldsymbol{\beta}$)	Rank($\boldsymbol{\beta}$)	$\overline{\sigma^2_{yo}}$	$\overline{\sigma^2_{yc}}$	p_k	$\overline{d}_\beta$	δ_1	δ_2
8	2.648	2.689	5	4.674	0.400	0.50	146.70	1.00	1.00
9	2.664	2.682	4	5.610	0.400	0.10	246.22	1.00	1.00
10	2.682	2.703	10	4.703	0.384	0.50	147.05	0.50	0.50
11	2.682	2.696	8	5.591	0.391	0.10	246.33	0.75	0.75
12	2.687	2.698	9	5.656	0.384	0.10	246.76	0.50	0.50
13	2.689	2.719	11	4.714	0.390	0.50	146.79	0.75	0.75
14	2.955	3.583	13	8.487	15.583	0.50	137.63	4.00	4.00
15	3.497	3.815	23	10.610	0.624	0.50	62.70	3.00	2.50
16	3.588	3.900	35	11.163	0.623	0.50	40.97	4.00	2.50
17	3.648	3.794	18	11.265	0.432	0.10	137.72	2.00	1.50
18	3.669	3.818	25	11.390	0.431	0.50	60.39	2.00	1.50
19	3.693	3.761	14	11.332	0.391	0.10	167.17	1.00	0.75
20	3.701	3.795	19	11.542	0.400	0.10	136.95	1.50	1.00

The results in Table 1 show that the proposed decomposition is most successful when the two fields are given comparable graph-frequency regularization. Equal-penalty ($\delta_1 = \delta_2$) runs dominate the upper portion of the table, although some asymmetric settings also appear among the top twenty. This pattern is informative. In this experiment, the induced model discrepancy is the error created by replacing spatially varying diffusivities in the data-generating advection–diffusion system by constant diffusivities in the computer model. Figure 3 shows that this error has broad zonal structure and coherent meridional gradients. Thus, both the latent field and the discrepancy contain important low-frequency components. If the prior strongly favors one field over the other through unequal roughness penalties, the allocation mechanism is forced to solve a physical separation problem under an artificial asymmetry. The equal-penalty cases avoid this additional source of regularization bias: the likelihood and the Bernoulli allocation variables determine which low- and medium-frequency modes should be attributed to $\boldsymbol{X}$ and which should be attributed to $\boldsymbol{\beta}$.

The lower-ranked fits in Table 1 illustrate the cost of such asymmetry. Within the top twenty fits, the asymmetric runs all have $\delta_1 > \delta_2$, so that the bias coefficients are more heavily penalized than the process coefficients at the same graph frequency. These settings produce RMSE($\boldsymbol{X}$) and RMSE($\boldsymbol{\beta}$) values that are substantially larger than those in the equal-penalty runs. The estimated observation variance $\overline{\sigma^2_{yo}}$ also increases sharply, from approximately 4.5–5.7 in the best equal-penalty cases to approximately 11.3–11.8 in the unequal-penalty cases. This is consistent with the structure of the hierarchical model. The ensemble likelihood strongly constrains the sum $\boldsymbol{X} + \boldsymbol{\beta}$, whereas the physical observations constrain only $\boldsymbol{X}$ at the sampled locations. When the prior over-shrinks the discrepancy field, part of the model error must be absorbed by the latent field. The resulting latent field is then less compatible with the physical observations, and the posterior can partially accommodate this conflict by inflating σ^2_{yo}. Apart from the large-variance rows 1, 3, and 14, $\overline{\sigma^2_{yc}}$ remains relatively stable across these fits, mostly between 0.38 and 0.49. This stability is expected because the within-ensemble sum of squares W^c enters the full conditional distribution for σ^2_{yc} directly and is not strongly affected by a particular allocation of the ensemble mean into $\boldsymbol{X}$ and $\boldsymbol{\beta}$.

The dependence on the Bernoulli allocation prior is more modest as we do not see the cases where $p_k = 0.1$ as being significantly better than $p_k = 0.5$. The difference in latent-field recovery is negligible, while the discrepancy recovery is slightly better when the prior places more mass on assigning eligible modes to $\boldsymbol{\beta}$. This should not be interpreted as a general preference for $p_k = 0.10$. Rather, it reflects the design of this particular experiment: the computer-model error in this case has appreciable low- and medium-frequency content, so allowing a larger number of eligible modes to enter the discrepancy field can improve recovery. The posterior mean of d_β also shows that our simulations are not simply reproducing the prior distributions. When $p_k = 0.10$, the prior mean number of bias modes among the $K_0 = 300$ eligible modes is about 270, yet the top equal-penalty fits have $\overline{d_\beta}$ near 246. The likelihood therefore pulls a nontrivial number of eligible modes back into the process field. When $p_k = 0.50$, $\bar{d}_\beta$ is near 146, close to but slightly below the prior mean of 150. Thus the prior controls the baseline size of the bias subspace, but the posterior allocation still reflects information in the observations and the ensemble mean.

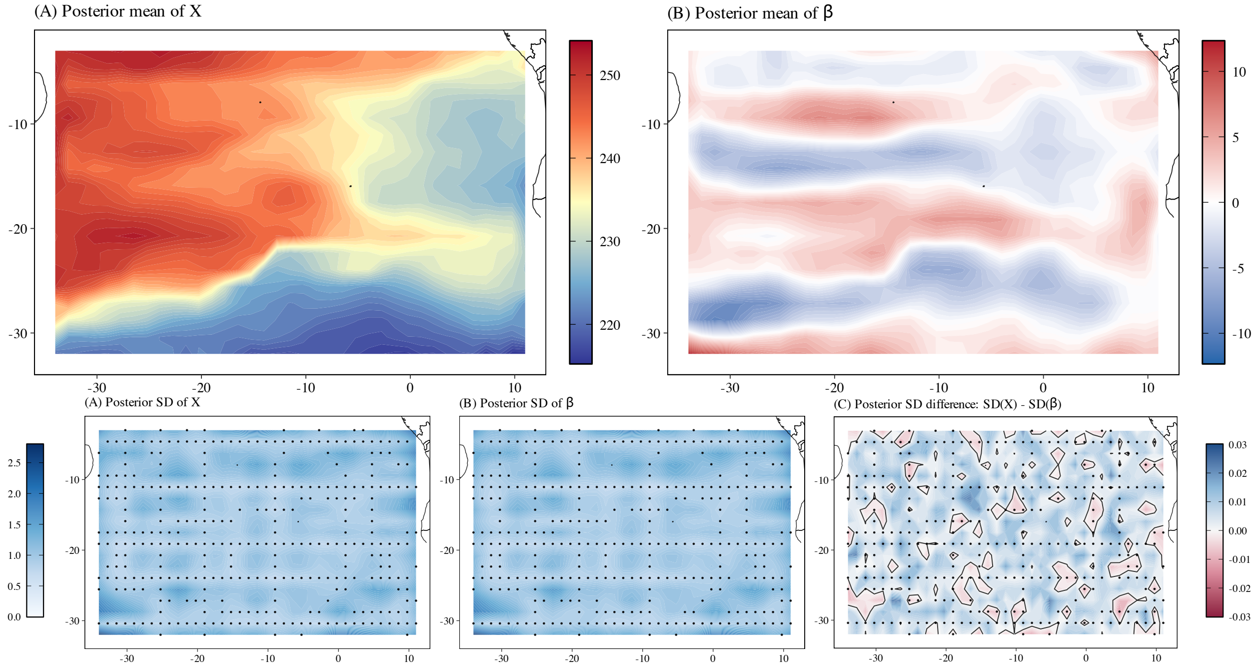

Figure 4: Posterior summaries under $\delta_1 = \delta_2 = 2.5$, $p_k = 0.5$. Top row: posterior means of the latent field $\boldsymbol{X}$ (left) and model bias $\boldsymbol{\beta}$ (right). Bottom row: posterior standard deviations for $\boldsymbol{X}, \boldsymbol{\beta}$ and the difference SD($\boldsymbol{X}$) - SD($\boldsymbol{\beta}$).

We also notice a close agreement between the rankings by RMSE($\boldsymbol{X}$) and by RMSE($\boldsymbol{\beta}$) and attribute this to the model geometry as well. When there is strong confidence in the computer model ensemble, it provides dense information about the additive mean $\boldsymbol{X} + \boldsymbol{\beta}$. Thus, an accurate estimate of $\boldsymbol{X}$ necessarily improves the implied estimate of the discrepancy $\overline{\boldsymbol{Y}}^c - \boldsymbol{X}$, and conversely a poor allocation of low-frequency discrepancy into $\boldsymbol{X}$ degrades both targets. The discrepancy is nevertheless slightly harder to estimate: it is not observed directly, but is inferred as the component of the ensemble mean that is inconsistent with the physical observations and the graph-frequency prior. The RMSE values in Table 1 are therefore consistent with the inferential roles of the two data sources.

Figure 4 provides a spatial view of row 2 in Table 1. The posterior mean of $\boldsymbol{X}$ reproduces the main features of the true latent field in Figure 3, including the large-scale west–east and north–south gradients. The posterior mean of $\boldsymbol{\beta}$ recovers the broad alternating bands visible in the induced discrepancy. Importantly, the fitted discrepancy is spatially organized and smooth, as one would expect from an error induced by misspecified transport diffusivities rather than by independent measurement noise. This behavior is the intended effect of restricting $\boldsymbol{\beta}$ to the low-frequency eligible modes and penalizing coefficients through the graph Laplacian spectrum. It is also consistent with the usual view in computer-model calibration that model discrepancy should be represented as a structured process rather than as an uncorrelated error term when the scientific mechanism suggests coherent bias.

The posterior standard deviation panels in Figure 4 show that uncertainty in $\boldsymbol{X}$ and $\boldsymbol{\beta}$ is of comparable magnitude under the selected simulation run, and the difference map SD($\boldsymbol{X}$) $-$ SD($\boldsymbol{\beta}$) is small across the lattice. This is not surprising in the equal-penalty case. Conditional on a modal allocation, the computer-model ensemble informs the spectral coefficients of both fields through the same dense ensemble mean, while the observations provide additional but spatially incomplete information about $\boldsymbol{X}$. The observation network improves local learning of the process field, but it does not remove the fundamental allocation uncertainty in the low-frequency subspace. The result is that the two posterior uncertainty surfaces are similar at the scale shown in the figure. This is a useful diagnostic: the method is not producing a bias estimate with artificially small uncertainty, nor is it treating the latent process as known once the ensemble mean is observed.

Figure 5 summarizes the remaining uncertainty in the modal decomposition. A posterior probability near one in the $\boldsymbol{X}$ row indicates that the corresponding graph-Laplacian mode is strongly assigned to the latent process; a high probability in the $\boldsymbol{\beta}$ row indicates strong assignment to the discrepancy field. The plot shows that the posterior allocation is not a simple cutoff in the mode index. Some very low-frequency modes are assigned to $\boldsymbol{X}$, while other low- and

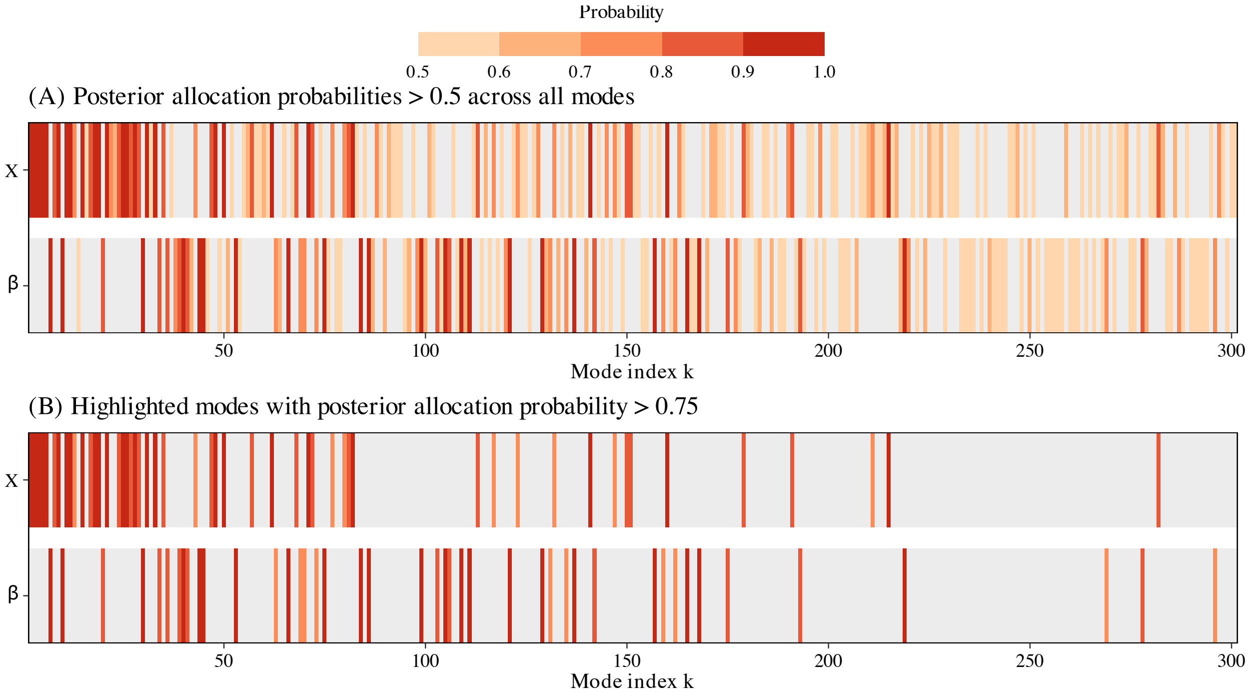

Figure 5: Posterior allocation probabilities under $\delta_1 = \delta_2 = 2.5$, $p_k = 0.5$. The $\boldsymbol{X}$ row displays $\mathbb{P}(B_k = 1 \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})$, and the $\boldsymbol{\beta}$ row displays $\mathbb{P}(B_k = 0 \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})$. The lower panel highlights modes whose posterior allocation probability exceeds 0.75.

medium-frequency modes are assigned to $\boldsymbol{\beta}$. This interleaving is scientifically sensible. The latent field itself is smooth and basin-scale, so the lowest frequencies cannot all be attributed to model bias. At the same time, the diffusivity-induced discrepancy also has smooth spatial organization, so it must occupy part of the same low-frequency region. The posterior allocation probabilities therefore encode the partially identified nature of the problem rather than conceal it. Modes with strong posterior probabilities correspond to spatial scales where the observations and ensemble mean provide enough evidence to distinguish process structure from model discrepancy; modes with weaker probabilities correspond to scales where the data do not decisively support one attribution over the other. In this respect, the allocation probabilities become an important inferential summary of which spatial scales are being attributed to the latent field and which are being attributed to structured model error, a distinction that is central in partially identified Bayesian discrepancy problems (Gustafson, 2015).

5. Data application

The data application concerns monthly mean sea-surface temperature (SST) over the Pacific Ocean on a fixed ocean lattice. We analyze the January 2010 monthly field. SST regulates air–sea heat exchange and strongly constrains large-scale climate variability, especially across the tropical Pacific, where basin-wide temperature gradients shape atmospheric circulation and coupled climate fluctuations (Deser, Alexander, Xie and Phillips, 2010). The purpose of the application is to combine an observation-based SST analysis with a climate-model ensemble and to estimate, on a common lattice, both a latent SST field $\boldsymbol{X}$ and a spatially structured climate-model bias field $\boldsymbol{\beta}$, as a real-world test of the feasibility of the methodology described above.

Observation-based sea-surface temperature analysis. For the observational component we use the monthly mean NOAA/NCEP Optimum Interpolation SST product, version 2, distributed in the file `sst.mnmean.nc`. This product combines ship, buoy, and satellite observations through optimum interpolation to provide global monthly fields on a $1^\circ \times 1^\circ$ latitude–longitude grid (Reynolds, Rayner, Smith, Stokes and Wang, 2002; Reynolds and Stokes, 1981; NOAA Physical Sciences Laboratory, 2026). The variable `sst` is reported in degrees Celsius. We restrict the global analysis to a Pacific-domain window and denote the resulting full observation-based field by $\boldsymbol{Y}^{o,\mathrm{full}} \in \mathbb{R}^S$. The likelihood uses only $N = 1000$ ocean cells sampled uniformly at random from this domain. If $\mathcal{O}_{\mathrm{obs}}$ denotes those sampled cells, then

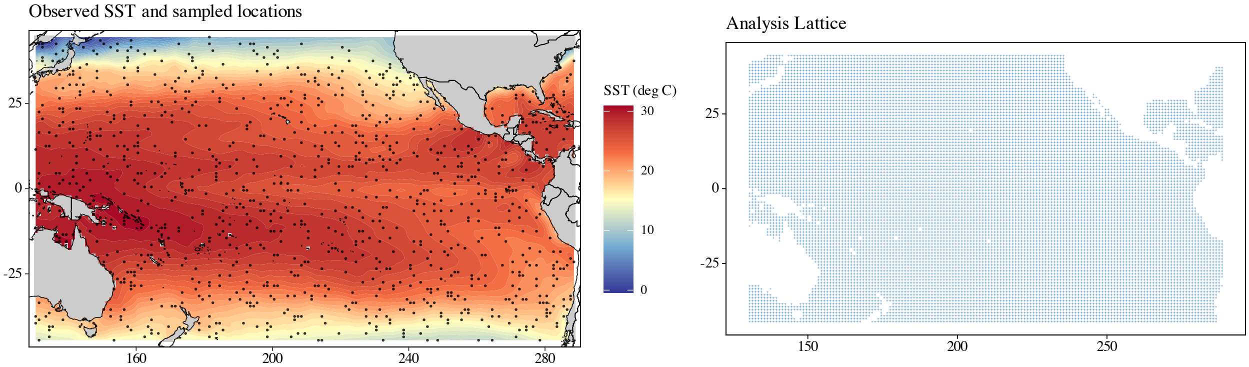

Figure 6: Observation-based monthly mean SST over the Pacific domain and the $N = 1000$ sampled locations used in the application (left). Analysis lattice used to define the graph-based prior distributions (right).

the observation vector in the model is $\boldsymbol{Y}^o = \boldsymbol{Y}^{o,\text{full}}_{\mathcal{O}_{\text{obs}}}$ and the sampling matrix $\mathcal{H}$ selects the same entries from $\boldsymbol{X}$. The left panel of Figure 6 shows the observation-based SST field together with the sampled locations.

Climate-model ensemble. For the computer-model component we use monthly mean SST from version 2.0 of the Energy Exascale Earth System Model for the historical experiment of the Coupled Model Intercomparison Project Phase 6 (Eyring, Bony, Meehl, Senior, Stevens, Stouffer and Taylor, 2016; Golaz, Van Roekel, Zheng, Roberts, Wolfe, Lin, Bradley, Tang, Maltrud, Forsyth et al., 2022; Bader, Leung, Taylor and McCoy, 2022; E3SM-Project, 2023). The relevant model variable is `tos`, defined in the file metadata as sea-surface temperature at the upper boundary of the liquid ocean, reported in degrees Celsius and averaged over ocean area and over the monthly interval (Bader et al., 2022; E3SM-Project, 2023). The historical files used here are provided on the standard regridded $1^\circ \times 1^\circ$ latitude–longitude grid, obtained from the native mesh by an area-preserving remapping procedure (Bader et al., 2022; E3SM-Project, 2023). We analyze $M = 21$ ensemble members, consistent with the documented E3SM version 2 large-ensemble configuration (Fasullo, Golaz, Caron, Rosenbloom, Meehl, Strand, Glanville, Stevenson, Molina, Shields, Zhang, Benedict, Wang and Bartoletti, 2024). After restriction to the Pacific domain, the common analysis lattice contains $S = 12\,554$ ocean cells. The right panel of Figure 6 displays this lattice, and we use the standard north–south–east–west neighborhood structure to define the graph Laplacian. The climate-model fields are denoted by $\boldsymbol{Y}^c_1, \dots, \boldsymbol{Y}^c_M \in \mathbb{R}^S$.

Figure 7 displays four representative ensemble members. Their dominant structure is a warm western and central tropical Pacific, a cooler eastern equatorial Pacific, and strong meridional cooling away from the tropics. The ensemble members differ locally, but those differences are small relative to the basin-scale SST gradient. Figure 8 shows the corresponding pointwise sample variance across the 21 members. The variance field is heterogeneous and right-skewed, with its largest values concentrated near the equatorial Pacific and along parts of the southern and eastern domain margins. Thus, the ensemble contains information about a smooth large-scale mean field while retaining geographically structured internal spread.

Spectral truncation and validation diagnostics. The lattice used in this application (see the right panel in Figure 6) is no longer regular, but it is connected. A one-time, costly ($\approx$ 15 minutes on a basic desktop computer) full eigen-decomposition of the $S \times S$ Laplacian was used. However, a full representation with all $S = 12\,554$ Laplacian modes is unnecessary for this application and would be computationally inefficient. The update of $\boldsymbol{x}$ requires repeated factorization of $\boldsymbol{P}_x$ in (30), and the collapsed indicator sweep scales with the number of retained modes. More importantly, the observation-based likelihood uses only $N = 1000$ sampled values, so the data contain limited direct information about the finest spatial scales. We therefore use a truncated low-frequency version of the construction in Section 3. In this section, K denotes the number of retained modes, and, like before, $K_0 < K$ is the number of modes that are eligible for allocation, with $\mathcal{K}_0 = \{2, 3, \dots, K_0 + 1\}$. Let $\boldsymbol{U}_K = (\boldsymbol{u}_1, \dots, \boldsymbol{u}_K)$ denote the first K Laplacian eigenvectors and let $\boldsymbol{P}_K = \boldsymbol{U}_K \boldsymbol{U}_K^\top$ be the corresponding orthogonal projection. The fields $\boldsymbol{X}$ and $\boldsymbol{\beta}$ are represented only in this retained subspace, with disjoint retained modes assigned to the two fields. Thus the deterministic orthogonality constraint remains exact:

$$\boldsymbol{U}_\beta^\top \boldsymbol{U}_X = 0, \qquad \boldsymbol{\beta}^\top \boldsymbol{X} = 0.$$

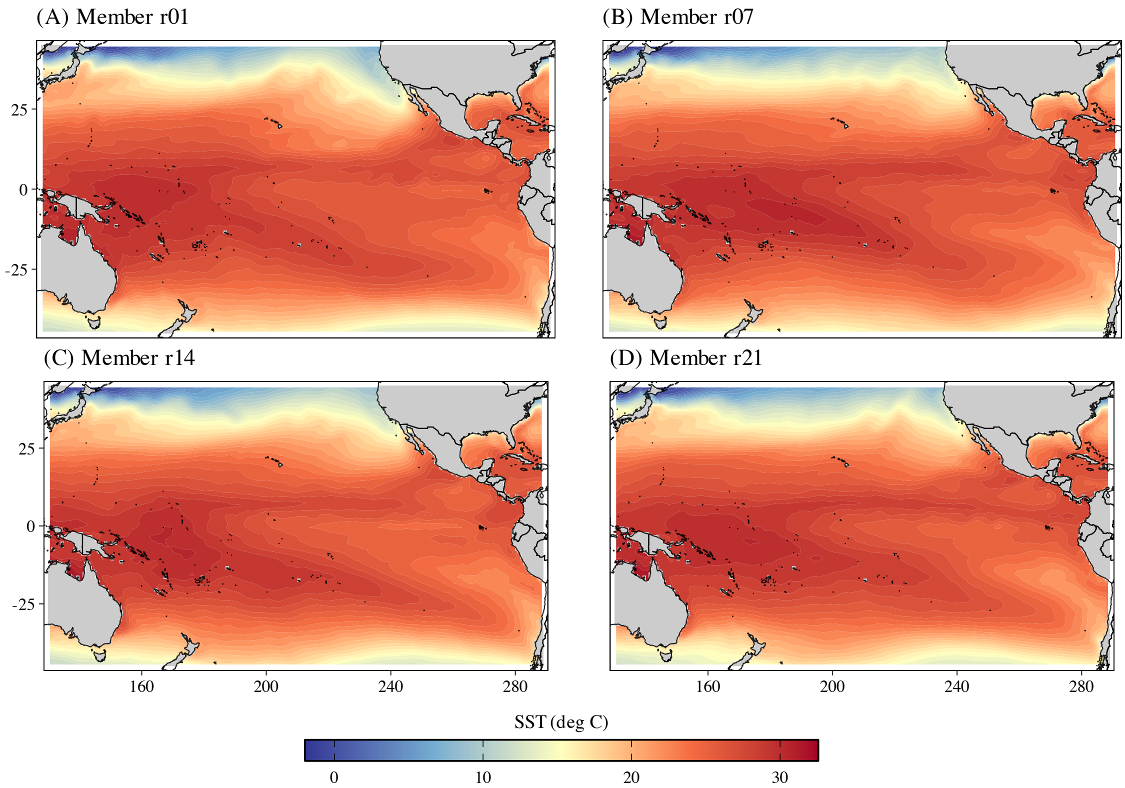

Figure 7: Four representative monthly mean SST ensemble members from the climate model.

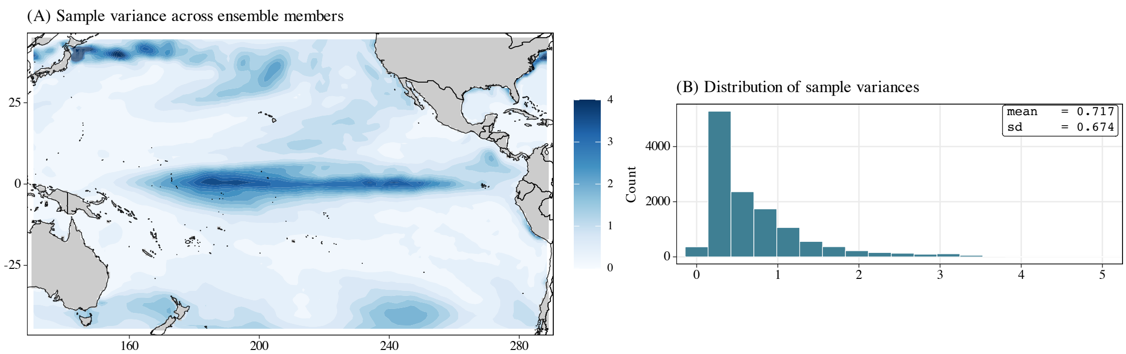

Figure 8: Pointwise sample variance across the 21 climate-model ensemble members (left) and histogram of those variances over the analysis lattice (right). The inset reports the mean and standard deviation of the lattice-wise variance field.

The difference from the full-basis construction is that

$$\boldsymbol{U}_{\beta}\boldsymbol{U}_{\beta}^{\top} + \boldsymbol{U}_{X}\boldsymbol{U}_{X}^{\top} = \boldsymbol{P}_{K}$$

rather than $\boldsymbol{I}_S$. Hence the reported fields should be interpreted as low-frequency reconstructions of the latent SST field and the model-bias field. The omitted component $\boldsymbol{P}_K^{\perp}\overline{\boldsymbol{Y}^c}$ is not assigned to either field; it is treated as residual ensemble variation. When the full spatial likelihood is used, this omitted component contributes the constant term

$$M\,\|\boldsymbol{P}_K^{\perp}\overline{\boldsymbol{Y}^c}\|^2$$

to the rate parameter in the update of σ^2_{yc}, but it does not affect the allocation probabilities or the Gaussian updates for the retained coefficients. To calibrate the truncation (select K), expand the ensemble mean in the full Laplacian basis as

$$\overline{\boldsymbol{Y}}^c = \sum_{k=1}^{S} y_k \boldsymbol{u}_k, \qquad y_k = \boldsymbol{u}_k^\top \overline{\boldsymbol{Y}}^c,$$

and define the cumulative fraction of Laplacian-basis energy captured by the first K modes,

$$R(K) = \frac{\sum_{k=1}^{K} y_k^2}{\sum_{k=1}^{S} y_k^2}.$$

The smallest values of K satisfying $R(K) \geq 0.90$, $R(K) \geq 0.95$, and $R(K) \geq 0.99$ are $K = 5$, $K = 12$, and $K = 20$, respectively. The ensemble mean is therefore extraordinarily smooth in the Laplacian basis. These very small values are not sufficient for the inferential objective, because both the latent SST field and the bias field may require additional low- and medium-frequency structure. As a conservative screening check, we also considered $K \in \{1000, 1200, 1500\}$, for which we computed the following summaries:

K	1000	1200	1500
$R(K)$	0.999909	0.999927	0.999942
RMSE(K)	0.2243	0.2019	0.1790

where RMSE(K) is computed on the analysis lattice after projecting $\overline{\boldsymbol{Y}}^c$ onto the first K Laplacian modes. These diagnostics indicate that the dominant SST structure resides at the smooth end of the spectrum. As a middle-ground choice, we take $K = 1200$ for the analysis. The MCMC runs below use $K_0 \in \{500, 700\}$ eligible nonconstant modes. These choices are far larger than the number of modes needed to capture the basin-scale ensemble mean, while still preventing the bias field from becoming an unrestricted high-frequency residual surface. We set $\delta_{01} = \delta_{02} = 10^{-4}$ and consider common roughness values (as suggested by the simulation study) $\delta_1 = \delta_2$ over the grid reported in Table 2.

For evaluation, we reserve observation-based SST values at hold-out cells not included in $\boldsymbol{Y}^o$. Let $\mathcal{O}_{\text{hold}}$ denote the hold-out set. Since the true latent field and true model bias are unknown, the following summaries are diagnostics rather than oracle errors:

$$\text{RMSE}(\boldsymbol{X}) = |\mathcal{O}_{\text{hold}}|^{-1/2}\, \|\widehat{\boldsymbol{X}}_{\mathcal{O}_{\text{hold}}} - \boldsymbol{Y}^{o,\text{full}}_{\mathcal{O}_{\text{hold}}}\|_2,$$

$$\text{RMSE}^{\text{proxy}}(\boldsymbol{\beta}) = |\mathcal{O}_{\text{hold}}|^{-1/2}\, \|\widehat{\boldsymbol{\beta}}_{\mathcal{O}_{\text{hold}}} - (\overline{\boldsymbol{Y}}^c_{\mathcal{O}_{\text{hold}}} - \boldsymbol{Y}^{o,\text{full}}_{\mathcal{O}_{\text{hold}}})\|_2.$$

The first diagnostic measures agreement between the posterior mean latent field and withheld values from the observation-based analysis. The second uses $\overline{\boldsymbol{Y}}^c - \boldsymbol{Y}^{o,\text{full}}$ as an empirical proxy for model bias; it should not be interpreted as validation against an observed bias field.

Table 2 shows that the common roughness value $\delta_1 = \delta_2$ is not the dominant tuning parameter in this application. For fixed K_0 and p_k, changing the common value of $\delta_1 = \delta_2$ produces only small changes in both hold-out diagnostics. This is consistent with the model structure: when $\delta_1 = \delta_2$ and $\delta_{01} = \delta_{02}$, a retained mode receives the same graph-frequency penalty whether it is assigned to $\boldsymbol{X}$ or to $\boldsymbol{\beta}$. The common roughness value regularizes both fields, but it does not create a relative preference for either component. The separation is instead driven primarily by the observation likelihood, the ensemble mean, the eligible modal range K_0, and the allocation prior p_k.

The more important sensitivity is the allocation prior. Recall that $p_k = \mathbb{P}(B_k = 1)$ is the prior probability that an eligible mode is assigned to the latent process. Thus $p_k = 0.1$ strongly favors assignment of smooth eligible modes to the bias field before the data are observed. In the SST application that prior is too bias-favoring. The observation-based analysis itself contains strong basin-scale structure, so allocating too many low-frequency modes to $\boldsymbol{\beta}$ leaves the latent process with too little flexibility to match withheld SST values. This explains why the $p_k = 0.1$ runs have larger hold-out errors and larger posterior means of σ^2_{yo}. By contrast, σ^2_{yc} remains essentially fixed near 0.727 across the table, reflecting the direct contribution of the within-ensemble sum of squares $\boldsymbol{W}^c$ in (32).

The posterior spatial summaries in Figure 9 use the first row of Table 2, namely $K_0 = 700$, $p_k = 0.5$, and $\delta_1 = \delta_2 = 4.0$. This run is tied for the smallest reported hold-out RMSE for $\boldsymbol{X}$ and is representative of the near-optimal $p_k = 0.5$ plateau. We do not interpret the table as strong evidence that $\delta = 4$ is intrinsically preferred; the tied second row differs only in the fourth decimal place of the proxy bias diagnostic. The scientific conclusion is instead that a neutral allocation prior with a sufficiently rich low-frequency candidate space gives a stable decomposition.

Table 2
MCMC results for the Pacific SST application. The columns report hold-out diagnostics for the posterior mean fields, posterior means of the variance components, the number K_0 of eligible nonconstant modes, the common prior allocation probability p_k, the posterior mean number of bias modes $\overline{d}_\beta$, and the common roughness setting $\delta_1 = \delta_2$. The second column is a proxy diagnostic based on $\overline{\boldsymbol{Y}}^c - \boldsymbol{Y}^{o,\mathrm{full}}$.

RMSE($\boldsymbol{X}$)	RMSE$^{\mathrm{proxy}}$($\boldsymbol{\beta}$)	$\overline{\sigma^2_{yo}}$	$\overline{\sigma^2_{yc}}$	K_0	p_k	$\overline{d}_\beta$	$\delta_1 = \delta_2$
1.2620	1.2633	1.455126	0.727905	700	0.5	311.80	4.0
1.2620	1.2629	1.458932	0.727086	700	0.5	310.79	0.5
1.2639	1.2659	1.445552	0.727684	500	0.5	213.42	4.0
1.2644	1.2660	1.453508	0.727157	500	0.5	213.39	1.5
1.2644	1.2662	1.454218	0.727422	500	0.5	212.51	3.0
1.2647	1.2660	1.450428	0.727030	500	0.5	212.93	1.0
1.2647	1.2663	1.448609	0.727268	500	0.5	213.54	2.0
1.2648	1.2659	1.457403	0.727442	700	0.5	310.98	3.0
1.2660	1.2671	1.456213	0.727268	700	0.5	309.36	1.5
1.2664	1.2673	1.455861	0.727112	700	0.5	310.32	2.0
1.2681	1.2689	1.459858	0.727066	700	0.5	311.61	1.0
1.2694	1.2702	1.452519	0.727018	500	0.5	213.48	0.5
1.3908	1.3841	1.602067	0.727763	500	0.1	386.43	4.0
1.3911	1.3859	1.600588	0.727089	500	0.1	387.27	1.5
1.3914	1.3860	1.598868	0.727143	500	0.1	386.47	2.0
1.3919	1.3870	1.601964	0.727084	500	0.1	386.38	0.5
1.3921	1.3871	1.596202	0.727176	500	0.1	386.88	1.0
1.3927	1.3867	1.602003	0.727467	500	0.1	387.31	3.0
1.3979	1.3931	1.629780	0.727013	700	0.1	565.33	0.5
1.3980	1.3898	1.629768	0.727599	700	0.1	565.29	4.0
1.3989	1.3937	1.631640	0.727098	700	0.1	565.43	1.0
1.4006	1.3932	1.627462	0.727369	700	0.1	565.64	3.0
1.4014	1.3954	1.628300	0.727288	700	0.1	565.93	1.5
1.4021	1.3955	1.632869	0.727143	700	0.1	565.93	2.0

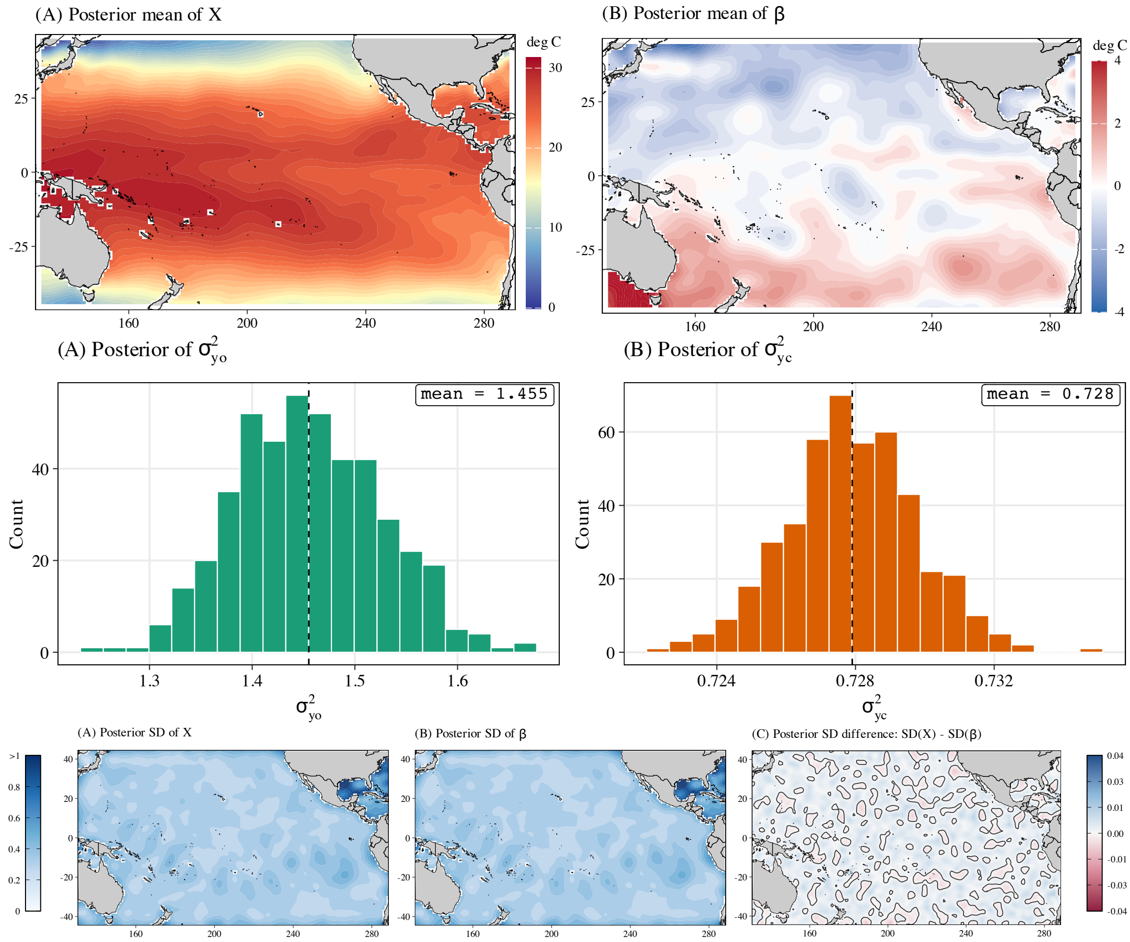

Figure 9: Posterior summaries for the SST application under $K_0 = 700$, $p_k = 0.5$, and $\delta_1 = \delta_2 = 4.0$. Top row: posterior mean latent SST field $\widehat{\boldsymbol{X}}$ and posterior mean bias field $\widehat{\boldsymbol{\beta}}$. Middle row: posterior histograms of σ^2_{yo} and σ^2_{yc}. Bottom row: posterior standard deviations for $\boldsymbol{X}$, $\boldsymbol{\beta}$, and the difference $\mathrm{SD}(\boldsymbol{X}) - \mathrm{SD}(\boldsymbol{\beta})$.

The posterior mean of $\boldsymbol{X}$ recovers the expected January Pacific SST organization: warm tropical and western Pacific waters, a cooler eastern equatorial region, and strong meridional cooling away from the tropics. The posterior mean of $\boldsymbol{\beta}$ is much smaller in amplitude than the latent field but is not negligible. Its dominant features are broad, geographically coherent sign changes across the basin rather than isolated grid-cell departures. Under the additive convention in (2), the posterior mean bias field identifies where the E3SM ensemble mean is inferred to be warm or cold relative to the latent SST field. Such basin-scale structure is scientifically plausible. Current coupled climate models are known to exhibit organized SST biases and large-scale discrepancies in Pacific SST patterns, rather than only spatially unstructured errors (Wills, Dong, Proistosecu, Armour and Battisti, 2022; Zhang, Liu, Li and Zhou, 2023).

The posterior standard deviation maps show similar marginal uncertainty for $\boldsymbol{X}$ and $\boldsymbol{\beta}$. As discussed in the simulation study, this is also a consequence of the decomposition problem, not a numerical artifact. The ensemble strongly informs the additive mean $\boldsymbol{X} + \boldsymbol{\beta}$, while the remaining uncertainty concerns how retained low-frequency modes are partitioned between the two components. Because the reported run uses symmetric graph-frequency shrinkage, the marginal uncertainty in the two fields is similar unless the observation likelihood is sufficiently strong to make the allocation nearly deterministic. The standard deviation difference map is therefore a useful diagnostic: the method is not treating the bias field as known once the ensemble mean is observed, nor is it producing an artificially certain latent SST field.

Figure 10 gives the main inferential summary of the modal decomposition. The allocation is not a simple low-frequency cutoff. Some very smooth modes are assigned with high probability to the latent field, while other low- and medium-frequency modes are assigned with high probability to the bias field. This interleaving is scientifically important. The latent SST field itself is a smooth basin-scale object, so assigning all of the lowest frequencies to model bias would be physically implausible. At the same time, a credible climate-model bias should be allowed to contain coherent large-scale structure. The posterior allocation probabilities express this balance directly: decisive probabilities indicate spatial scales where the data support an attribution, whereas probabilities closer to 0.5 indicate scales where the process–bias split remains partially identified.

Taken together, the SST results support the central modeling premise. The data do not justify an unrestricted residual surface for climate-model bias, but they do support a spatially coherent low-frequency correction whose sign and amplitude vary across the Pacific basin. The graph-Laplacian allocation model makes that correction estimable without pretending that the decomposition is uniquely determined by the likelihood alone.

6. Discussion

This paper develops a Bayesian modal-allocation approach for separating a latent spatial process from a structured model-bias field when a computer-model ensemble informs only the additive mean $\boldsymbol{X} + \boldsymbol{\beta}$. The central problem is identification. Without additional structure, many pairs $(\boldsymbol{X}, \boldsymbol{\beta})$ are compatible with the same ensemble mean, especially when physical observations are available only at a subset of spatial locations. The proposed construction replaces this continuous non-identifiability by a finite allocation problem in graph-frequency coordinates. The posterior distribution of the allocation indicators then becomes an interpretable scientific summary: it reports which spatial scales are attributed to the latent process and which are attributed to systematic model discrepancy.

The simulation study and the Pacific SST application illustrate complementary aspects of this mechanism. In the synthetic experiment, the true latent field and the induced discrepancy are available, and the best-performing runs occur when the process and bias fields receive comparable graph-frequency regularization. This is consistent with the design of the experiment: both the latent ocean field and the diffusivity-induced discrepancy have important low-frequency structure. Strongly asymmetric roughness penalties impose an artificial prior preference on a separation that should be learned from the observations and the ensemble mean. In the SST application, the true $\boldsymbol{X}$ and $\boldsymbol{\beta}$ are unavailable, so the RMSE summaries are hold-out and proxy diagnostics rather than oracle errors. Even so, the same qualitative lesson appears. Neutral allocation produces stable prediction, while a strongly bias-favoring allocation prior assigns too many smooth modes to $\boldsymbol{\beta}$, worsens prediction of the observation-based analysis, and inflates the observation-side variance. The posterior mean latent field retains the known large-scale structure of Pacific SST, while the posterior mean bias field is smoother, smaller in amplitude, and organized at basin scale. Thus, we interpret this as identification of a coherent spatial correction to the model mean. At the same time, the allocation probabilities show that the separation is not complete at every scale. Some modes are decisively attributed to either $\boldsymbol{X}$ or $\boldsymbol{\beta}$, whereas others remain close to prior neutrality. The method therefore avoids presenting an overconfident "true" bias map in a partially identified setting.

Several limitations remain. First, the decomposition is model-based. Orthogonality in the Laplacian basis is a defensible way to regularize the additive confounding, but it is not a prior-free discovery of a unique true bias field.

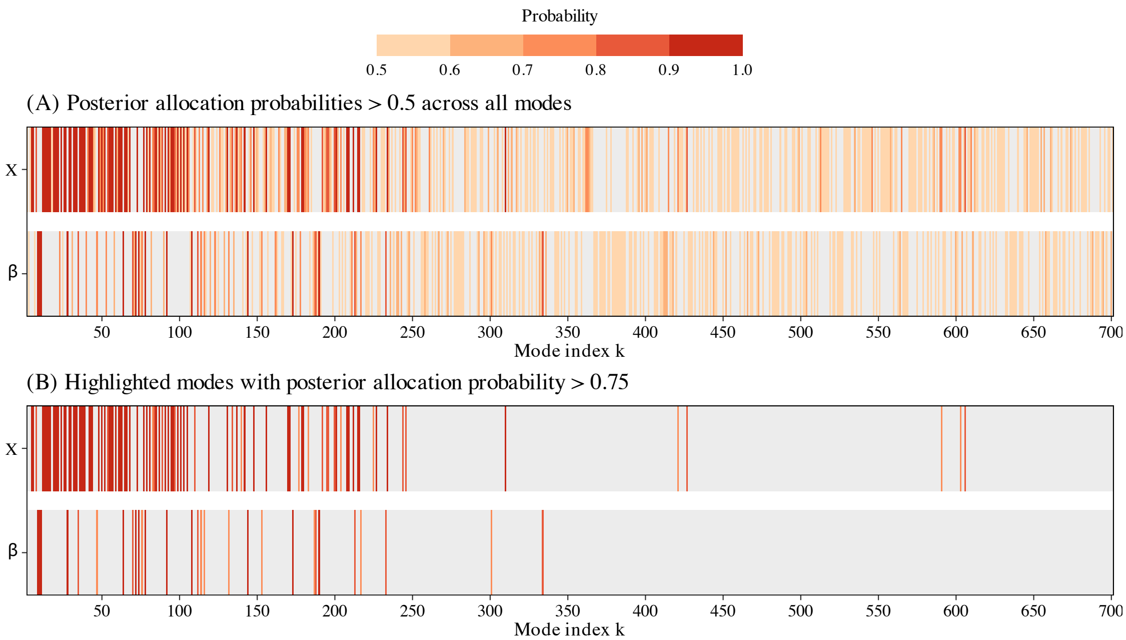

Figure 10: Posterior allocation probabilities for the SST application under $K_0 = 700$, $p_k = 0.5$, and $\delta_1 = \delta_2 = 4.0$. The X row displays $\mathbb{P}(B_k = 1 \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})$, and the β row displays $\mathbb{P}(B_k = 0 \mid \boldsymbol{Y}^o, \boldsymbol{Y}^c_{1:M})$. The lower panel highlights modes whose posterior allocation probability exceeds 0.75.

Applications in which discrepancy is localized, anisotropic, boundary-driven, or concentrated at unresolved fine scales may require different graphs, anisotropic weights, or richer allocation priors. Second, the SST validation uses an observation-based gridded product. Such products are essential for large-scale comparison, but their intercomparison depends on the reference data, grid, interpolation, and quality-control choices (Huang, Yin, Carton, Chen, Graham, Liu, Smith and Zhang, 2023). The hold-out diagnostics therefore measure agreement with a particular analysis product, not direct recovery of an unobserved physical truth. Third, the truncation level K_0 and allocation probabilities encode scientific assumptions about the spatial scales at which bias is allowed to operate. The results above suggest that prior-neutral allocation and symmetric roughness are useful defaults, but applications with stronger prior knowledge should encode that information explicitly.

Natural extensions include hierarchical models for the allocation probabilities, frequency-dependent logistic allocation priors, and hyperpriors on the roughness parameters. Spatio-temporal versions would allow persistent discrepancy patterns and evolving latent fields to be learned jointly, which is particularly important for climate applications. Multi-model extensions could distinguish shared structural bias from model-specific departures. On the computational side, larger and more irregular domains will require sparse eigensolvers, randomized spectral approximations, or transform-based implementations that avoid repeated dense factorizations.

The modal allocation described here provides a direct Bayesian route to a difficult decomposition problem. It preserves the interpretation of model discrepancy as a structured spatial field, avoids treating the computer model as unbiased truth, and summarizes the remaining non-identifiability through posterior probabilities over spatial scales. These features are useful when the scientific question is not only prediction of the latent field, but also attribution of coherent spatial structure to either process variation or model bias.

Data and computer code availability

Computer code used in this manuscript is available at `https://github.com/herbei/Bayesian_Allocation_Paper`. Information about how to download SST data is available there. In the figures above, map lines delineate study areas and do not necessarily depict accepted national boundaries.

Declaration of competing interest

The authors have nothing to declare.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI's ChatGPT/Codex to assist with grammar and spelling edits, reference checking, and debugging of LaTeX and R code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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